Optical Connector Contamination and its Influence on Optical Signal Performance

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Presentation Outline

- Introduction
- Experimental Methodology
- Effect of the scratches on IL & RL measurements;
- The influence of the particles on optical performance;
- Oil contamination and its effect on IL & RL measurements;
- BERT results for clean/contaminated fibers
- The prevention of the ESC effect during the cleaning process of fiber optics connectors
- Recommendations/Conclusions

Introduction

- The cleanliness of fiber optics connectors is recognized as the industry wide-problem;
- There is no industry-standard for cleanliness of fiber optics connectors;
- The cleaning and inspection processes of contaminated connectors affected the manufacturing time and therefore increase the test and product cost for high density port systems;
- The influence of the scratches/particles/oil contamination on optical performance was not investigated in details

- Object:
  - Learn the effects that many anomalies have on the performance of a fiber optic signal
  - Identify the severity of optical signal loss due to the most common, potential, hazards found in the supplier and internal manufacturing processes
- Scope:
  - Develop connector end-face inspection criteria, which would provide feedback to OEM Incoming Quality, Component Engineering and cable suppliers on specific cleanliness requirements with supporting data
  - Provide quantitative evidence to production and test departments to validate their expensive inspection and cleaning strategies, which have been historically endorsed by the industry

Fiber Optic Signal Performance Project

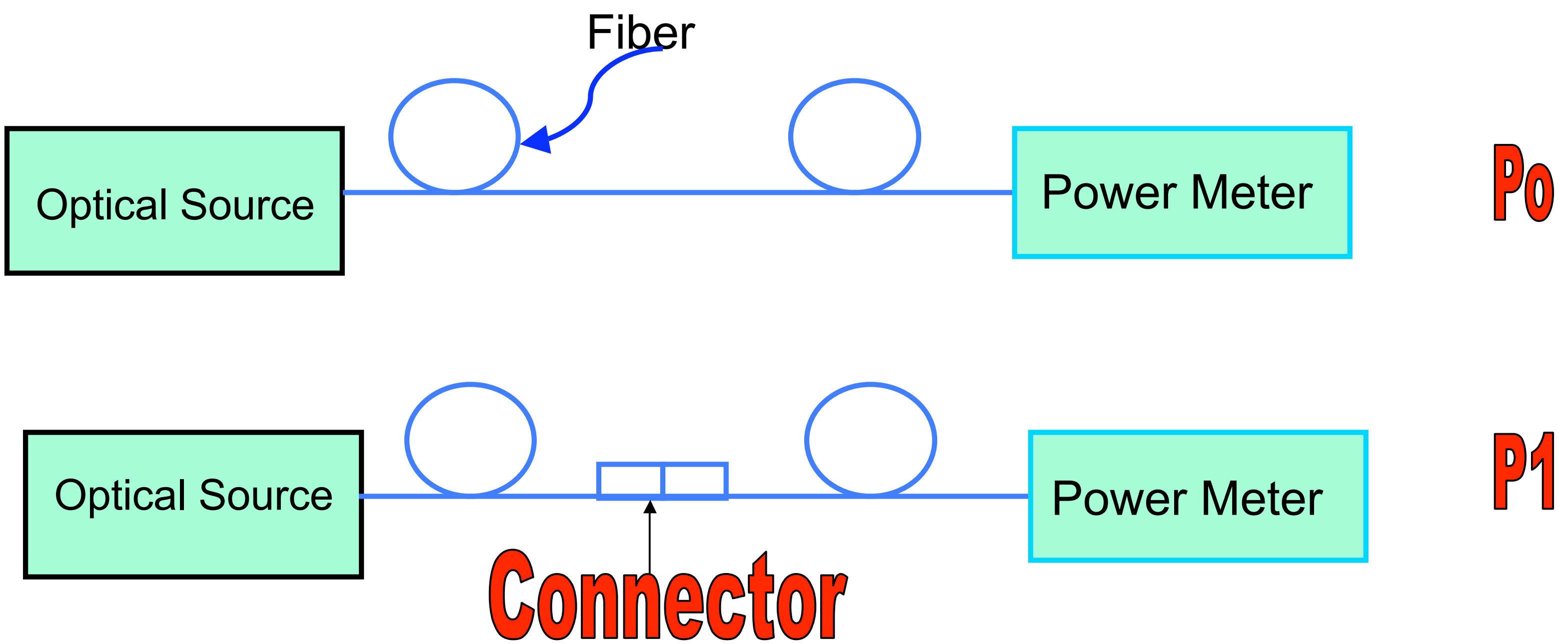
Benefits:

- Develop the Industry Standard for cleanliness of fiber optics connectors. Methods and guidelines was submitted for incorporation into OE standards, e.g. IPC-STD-040
- Improve the cleaning process and and prevent of fiber endface contamination

Project Status:

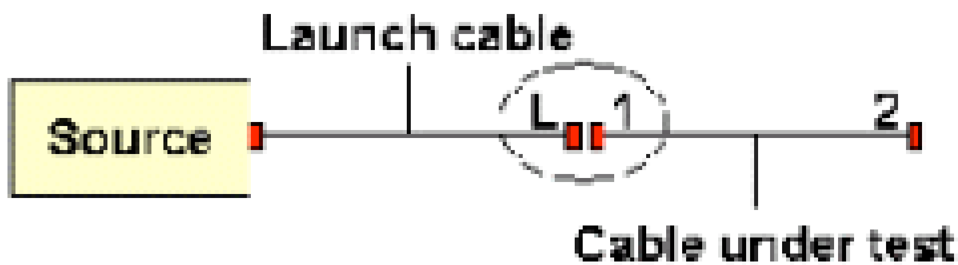
- First face-to face meeting held June 2002 (Celestica, Toronto)
- Confirmed participants:
  Alcatel Canada, Celestica, Cisco Systems, Nextron SA, Nortel Networks, Sanmina-SCI, Solectron, Aerotech
- The results of the the project has been presented at OMI conference (Ottawa, Apr, 03) and has been published at the Journal of SMT (2003, v.16, issue 3)
- Looking for additional members, particularly cable suppliers

Experimental Methodology - Insertion Loss (IL)



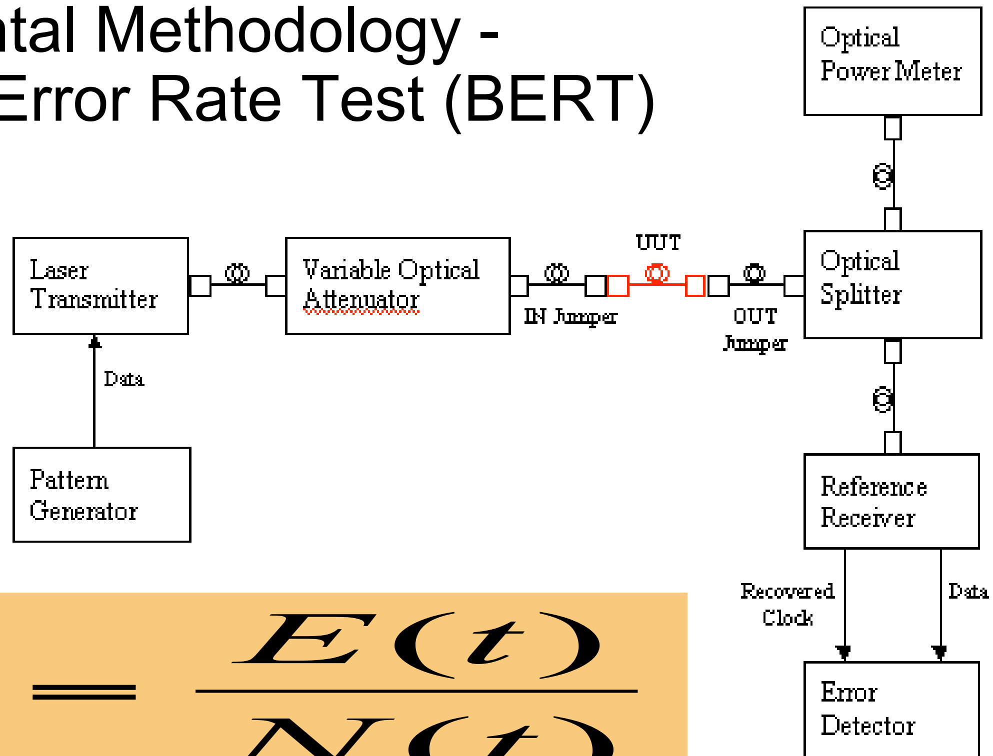
- **Insertion Loss =- 10 Log (P1/Po), dB**

Experimental Methodology - Return Loss (RL)



- Reflectance = 10 log Preflected, Pr/Pincident [dB]
- Return Loss= 10 log Pincident, Pi/ Preflected, Pr [dB]
- Return Loss = - Reflectance

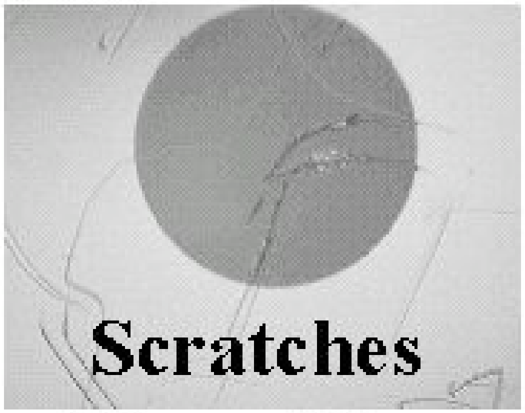
Experimental Methodology - Bit Error Rate Test (BERT)

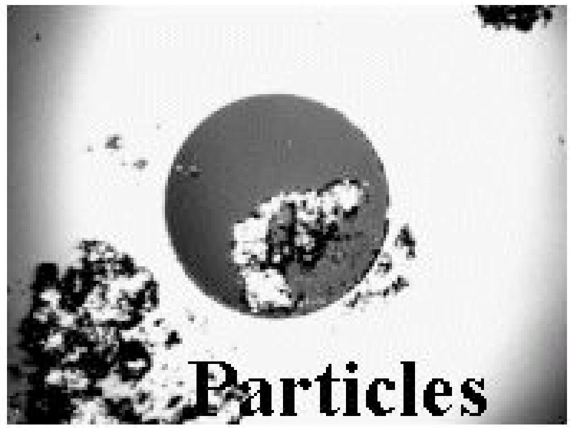


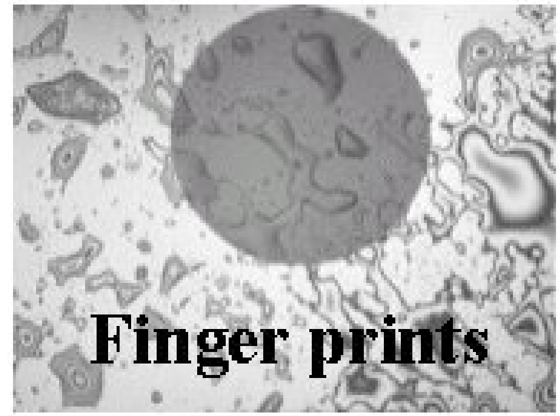
$$BER = \frac{E(t)}{N(t)}$$

Where *BER* is the bit error ratio
***E (t)* is the number of bits received in error over time *t*,**
and *N (t)* is the total number of bits transmitted in time *t*

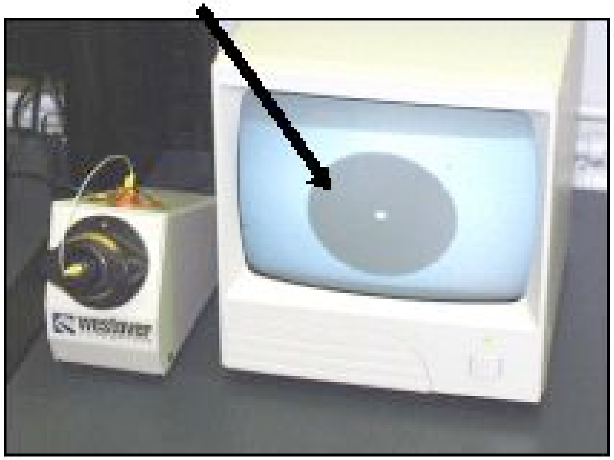
Fiber Optic Signal Performance Project





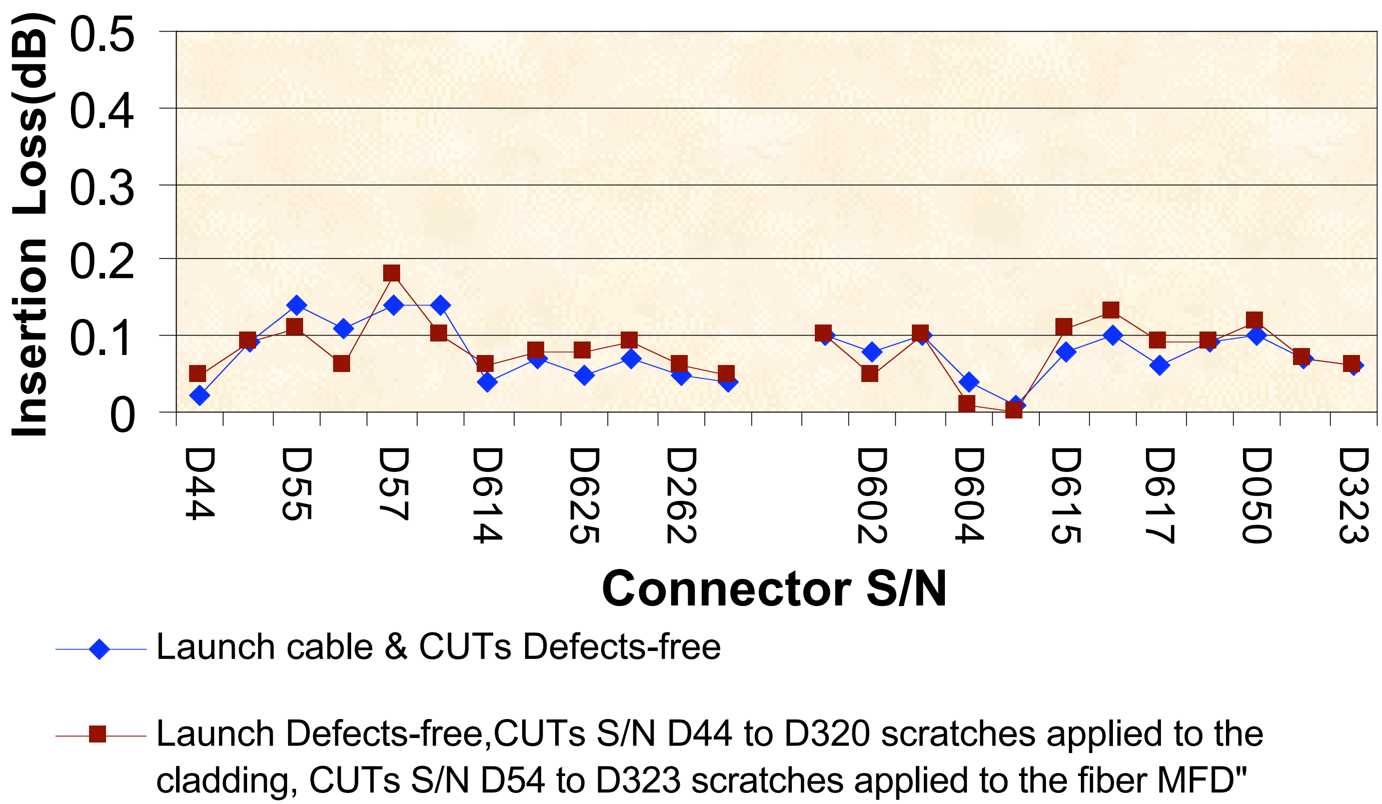


Clean Fiber

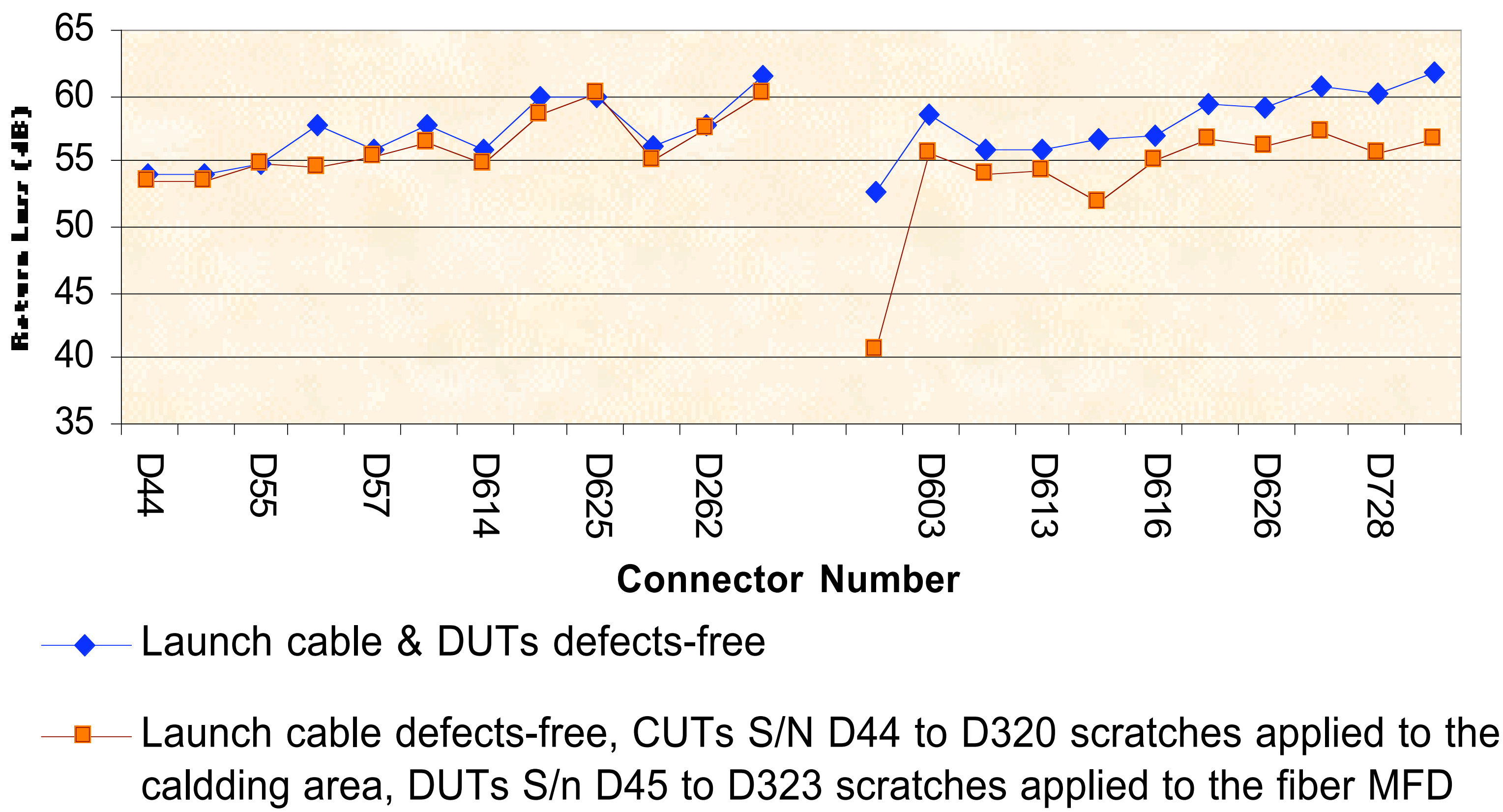


- Polishing scratches
- Particles
- Oil contamination (finger prints)

Effect of Scratches on Insertion Loss

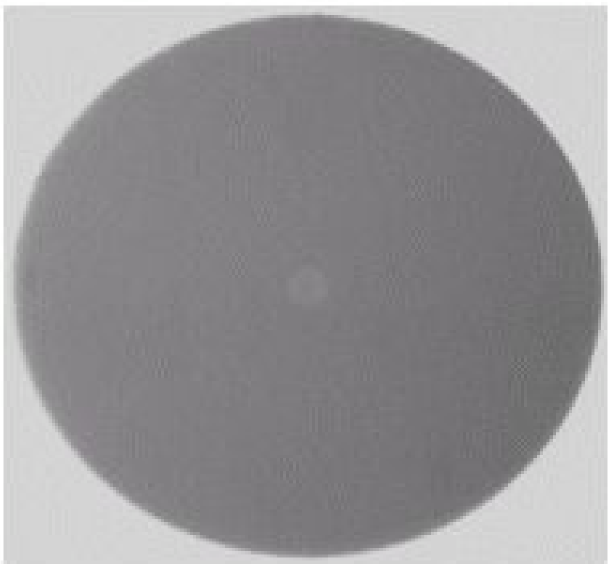


Effect of Scratches on Return Loss

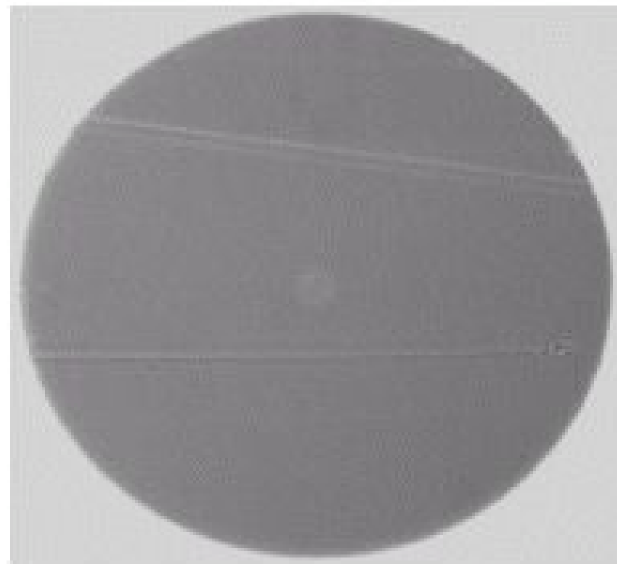


Effect of Scratches Through Cladding on IL/RL

FC # 55 Max IL: 0.14 dB;
RL: 54.7 dB

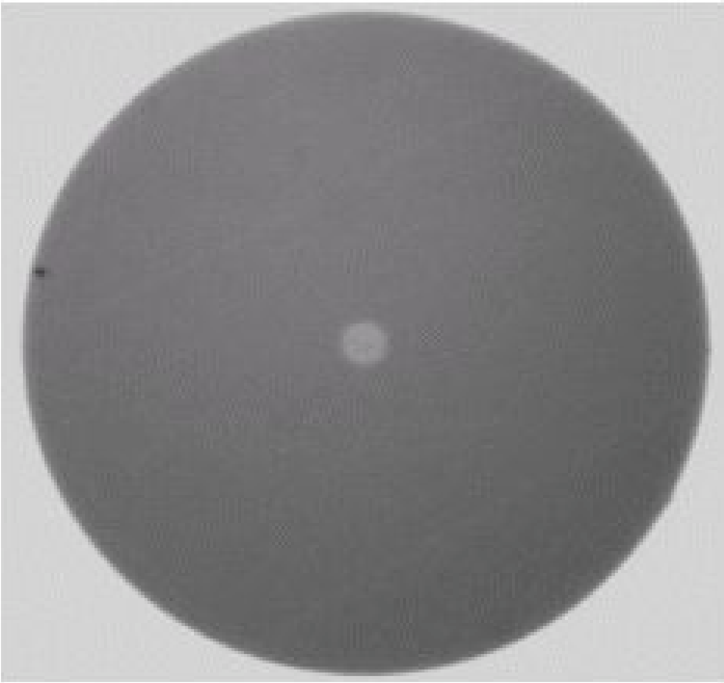


FC # 55 Max IL: 0.11dB;
RL: 54.8 dB

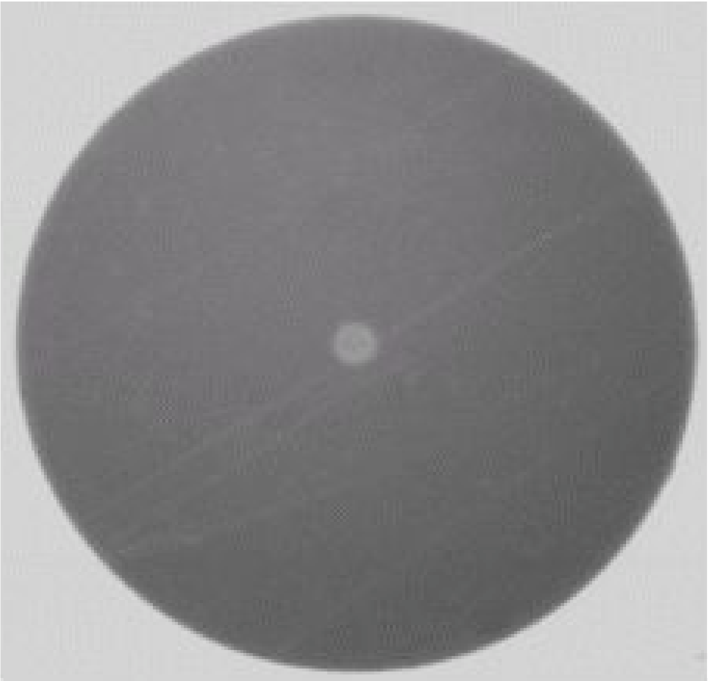


Effect of Scratches Through Core on IL/RL

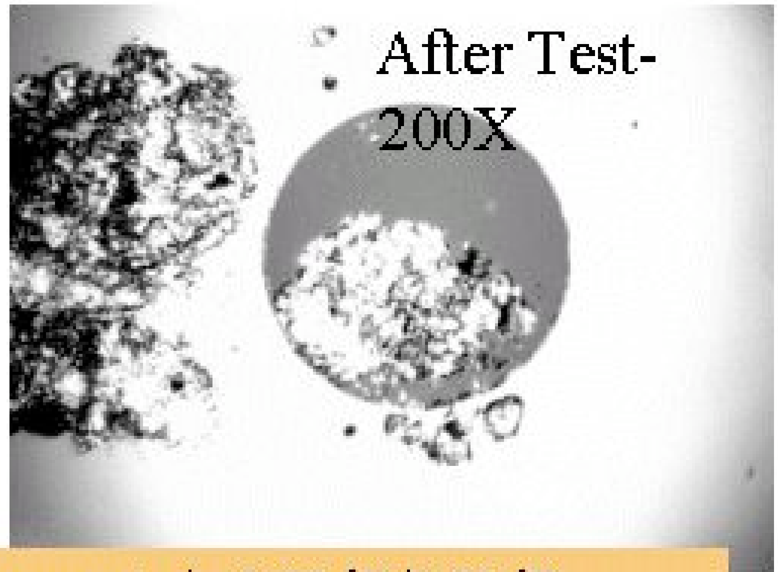
FC# 626 Max IL: 0.09dB; RL: 59.2 dB



FC# 626 Max IL: 0.09dB; RL: 56.2 dB



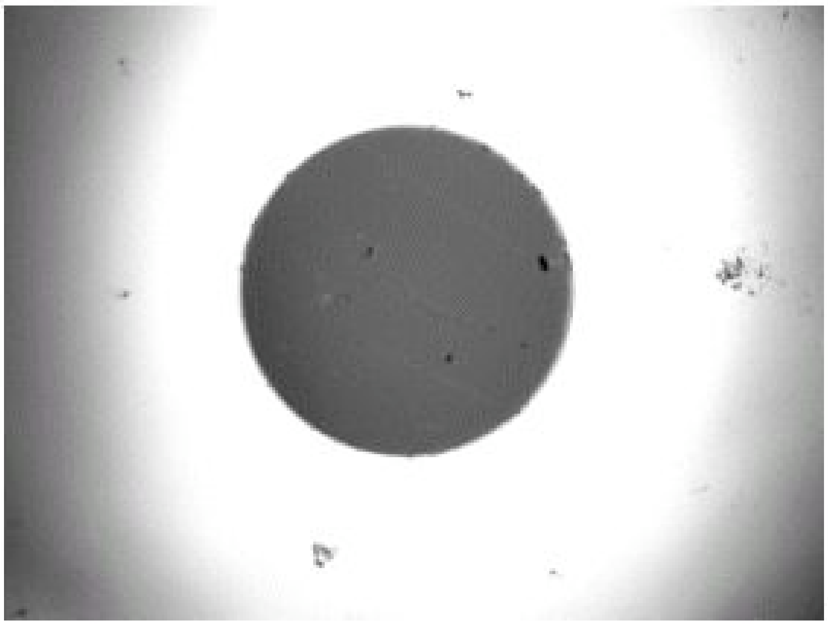
Particles on Core / Cladding / Ferrule



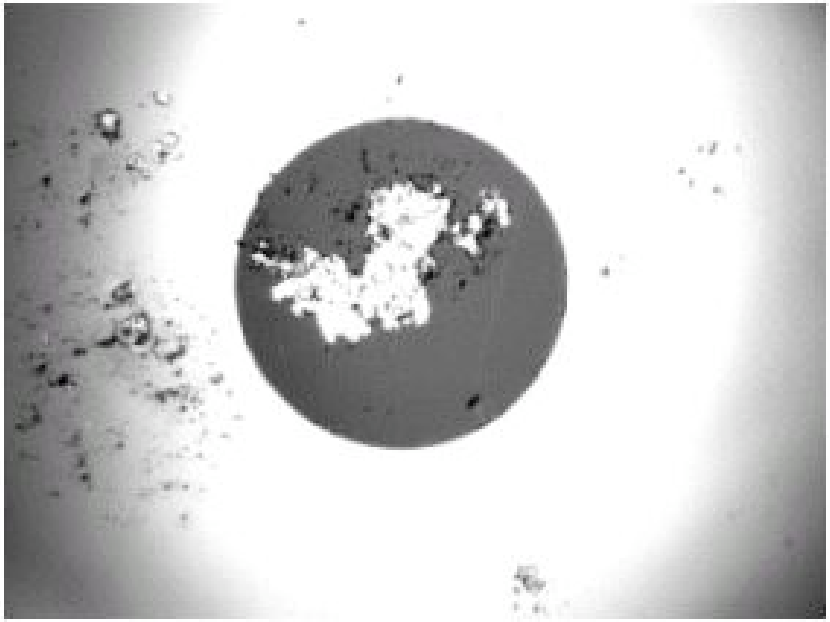
- IL-1550nm/1310nm (clean connector)=0.23dB/0.20dB;
- IL-1550nm/1310nm (contaminated)=21.9dB/22.8dB;
- RL-1550nm/1310nm(clean connector)=52.9dB/53.2dB;
- RL-1550nm/1310nm(contaminated)=20dB/20dB

Particles on Core / Cladding / Ferrule

- Test fiber becomes contaminated after the test

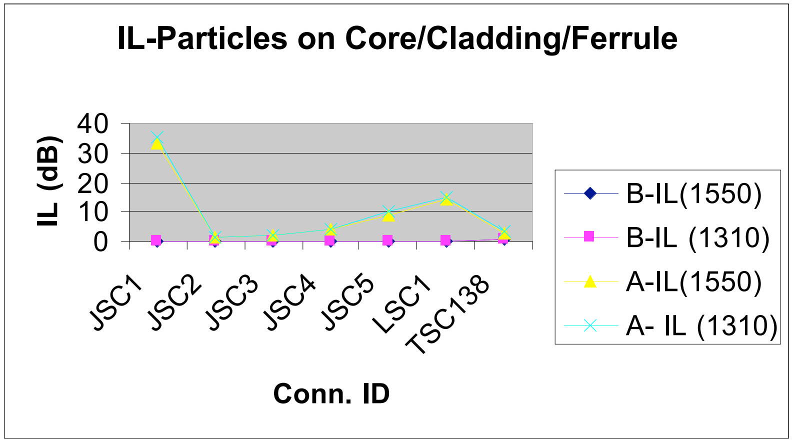


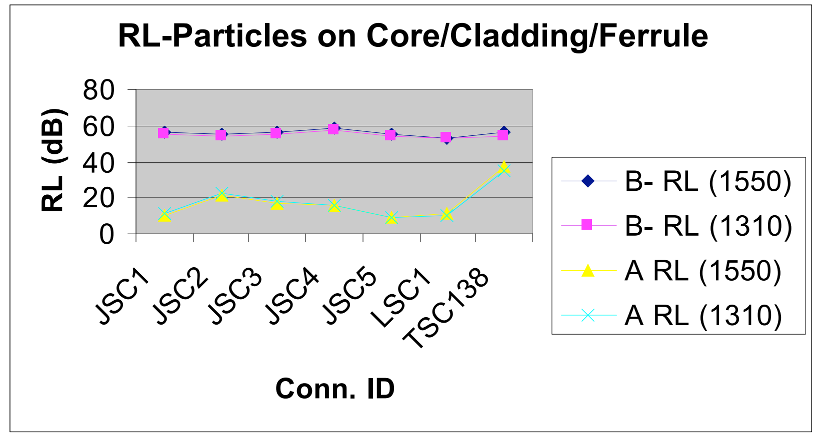
REFSC- Before Test

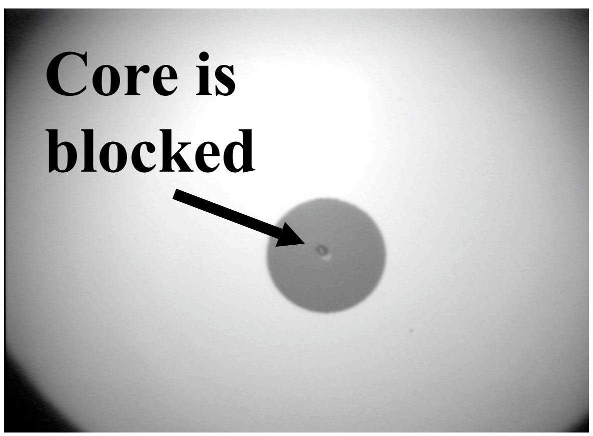


REFSC-After Test

Particles on Core / Cladding / Ferrule

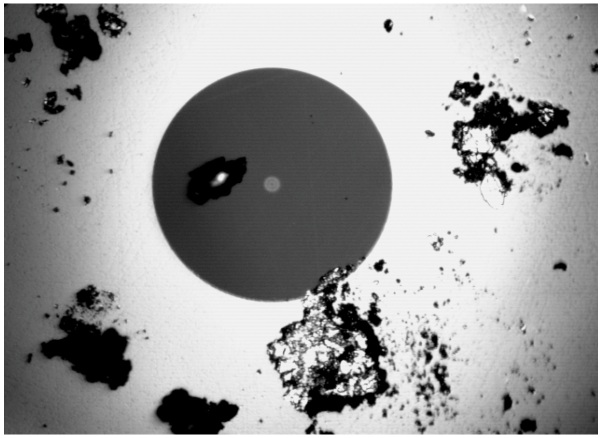




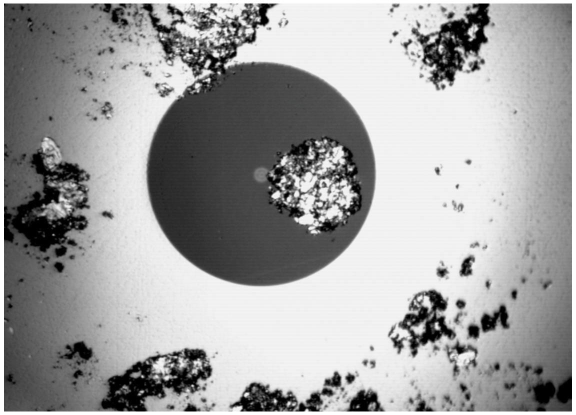


- IL-1550nm/1310nm(clean)=0.39/0.51dB
- IL-1550nm/1310nm(contaminated)=2.88/3.61dB
- RL-1550nm/1310nm(clean)=56.2/54.6dB;
- RL-1550nm/1310nm(contaminated)=37.1/34.5dB

Particles on Cladding / Ferrule Area



Before Test (Connection)



After Test (Connection)

IL-1550nm/1310 nm(clean connector)=0.14/0.15 dB,

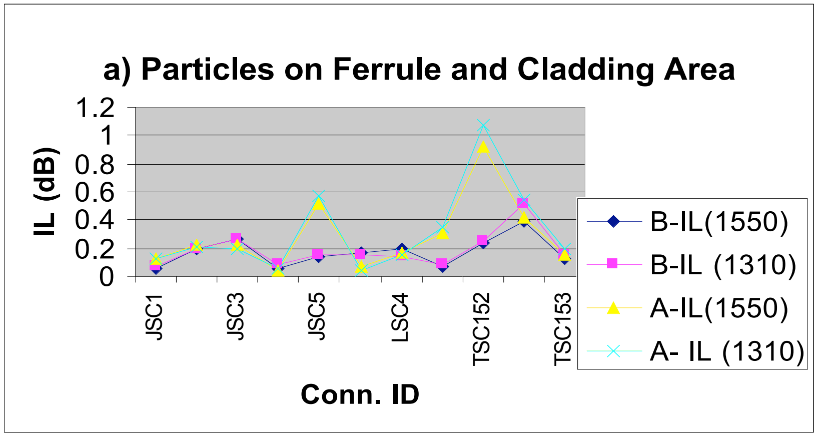
IL -1550nm/1310 nm (contaminated connector)=0.52/0.57 dB,

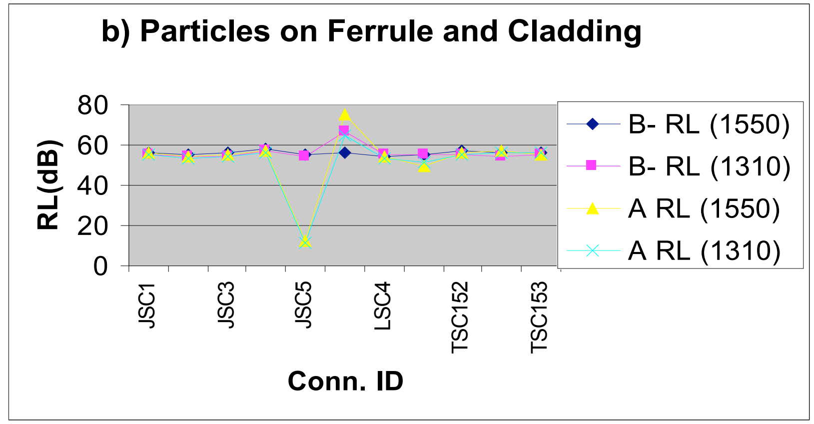
RL-1550/1310 nm (clean connector)=55.6/54.5 dB,

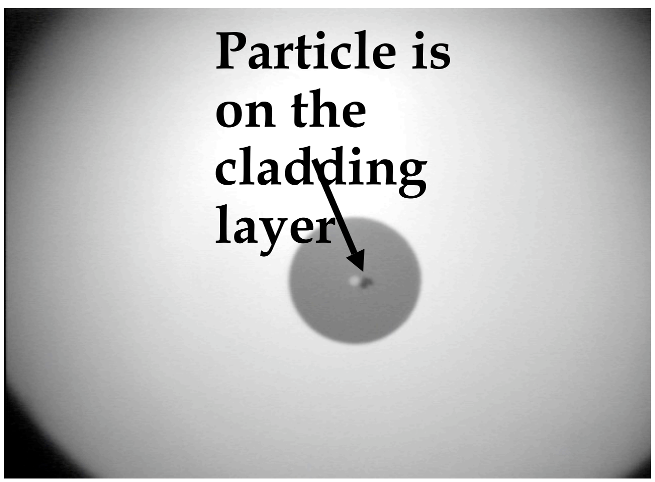
RL-1550nm/1310 nm (contaminated connector)=12.7/11.2 dB

Dave Silmser, Alcatel Canada, Inc. / Tatiana Berdinskikh, Celestica

Particles on Cladding / Ferrule

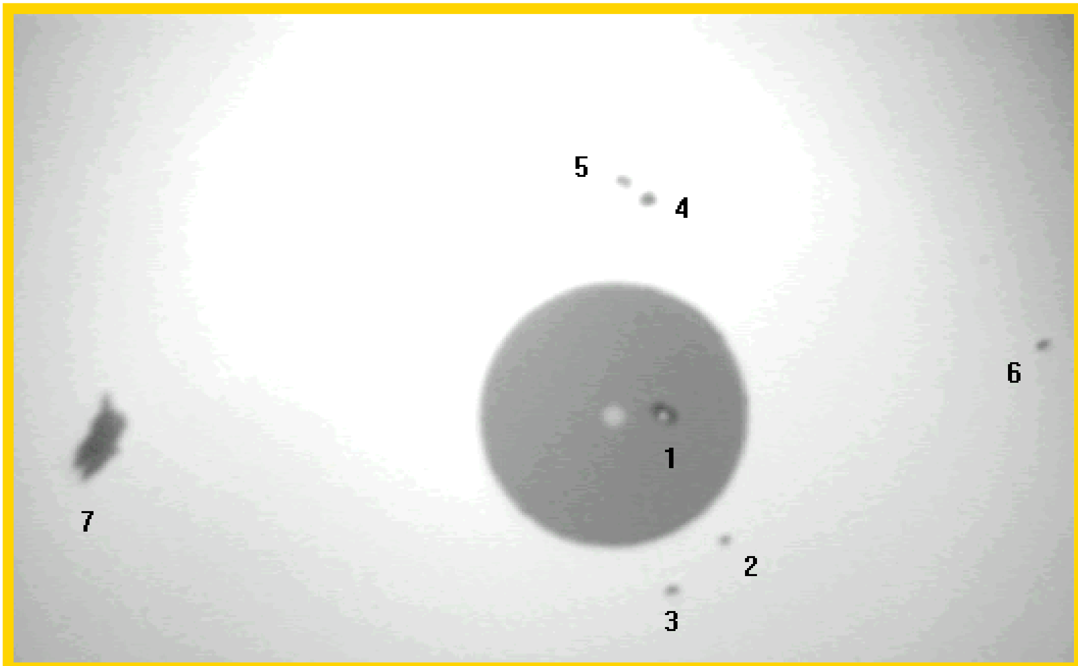






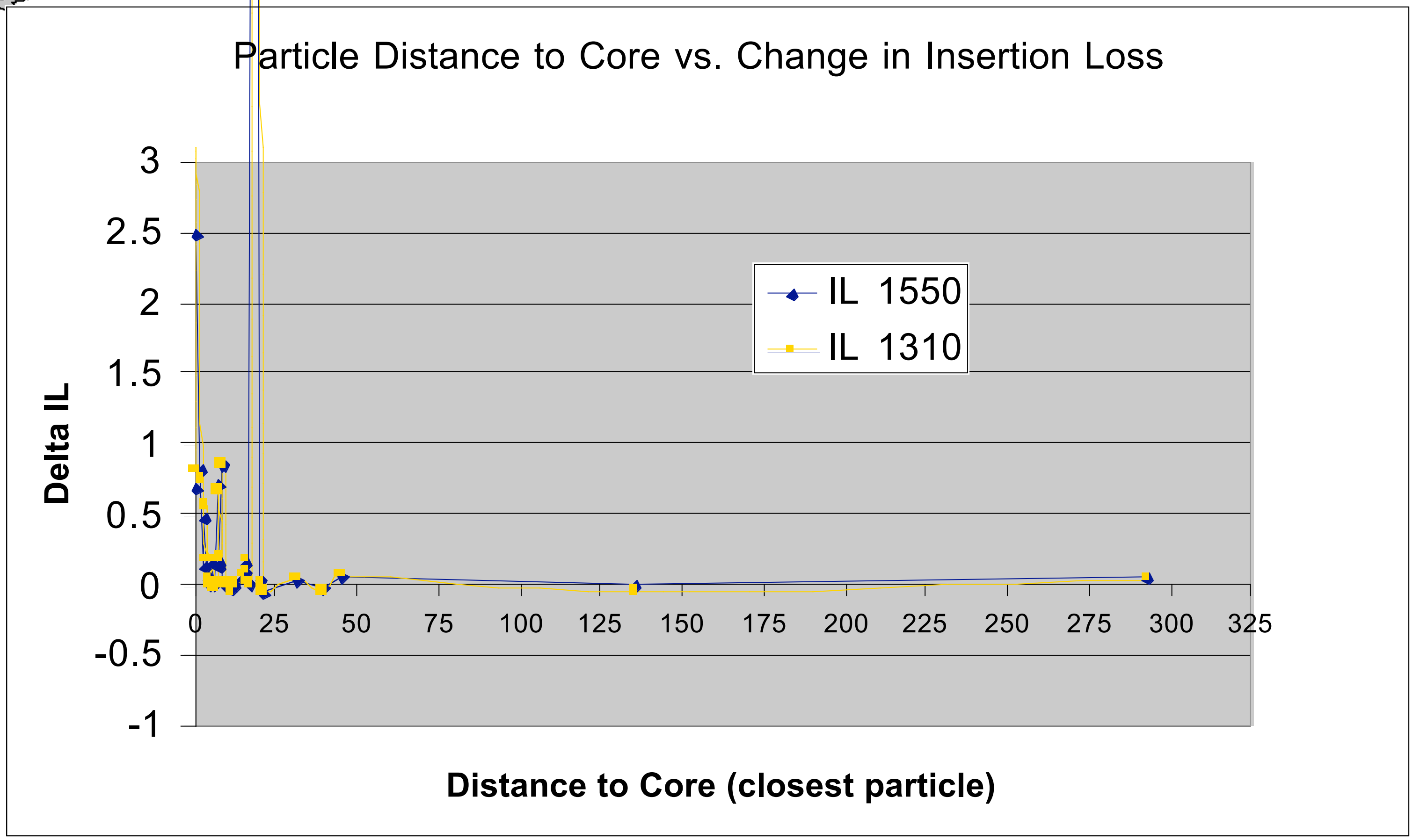
- IL-1550nm/1310nm(clean)=0.24/0.25dB;
- IL-1550nm/1310nm(contaminated)=0.92/1.07dB;
- RL-1550nm/1310nm (clean)=57.3/55.5dB;
- RL-1550/1310nm (contaminated)=56.5dB/55.6dB

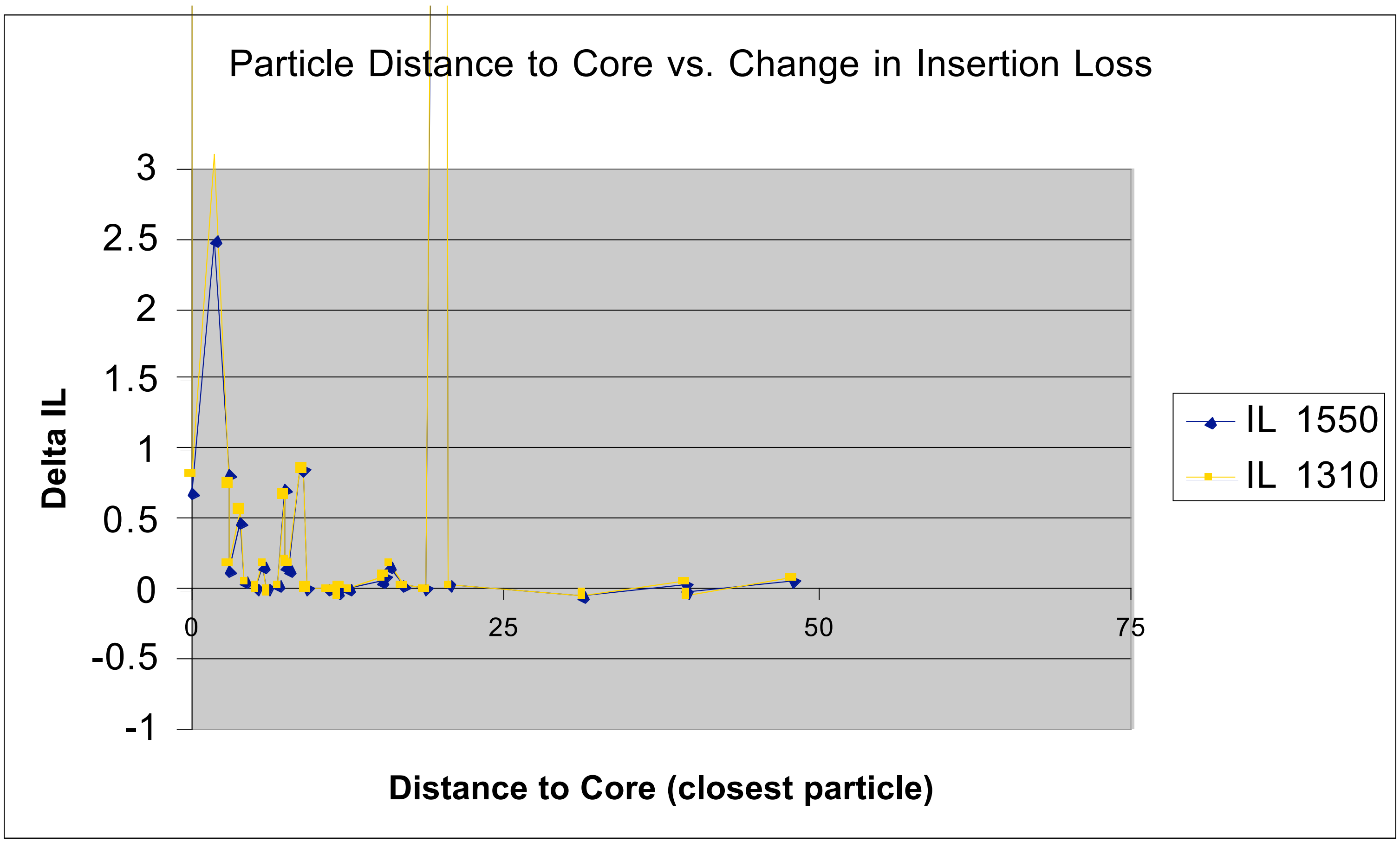
Fiber Optic signal Performance Project



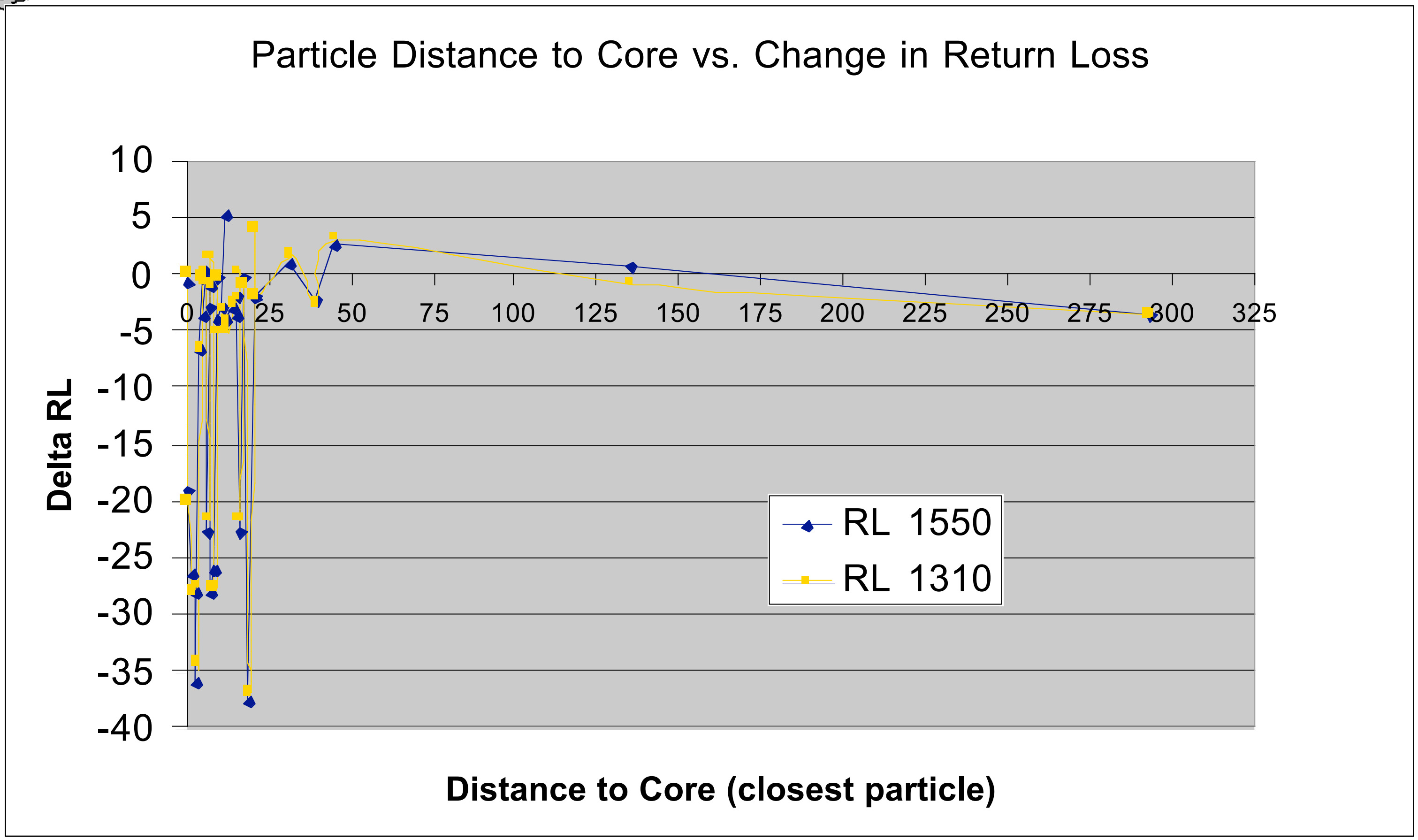
Measurement field #116.9micronMe	asurement field #277.4	micronMeasurement field #386.9micronMeasureme

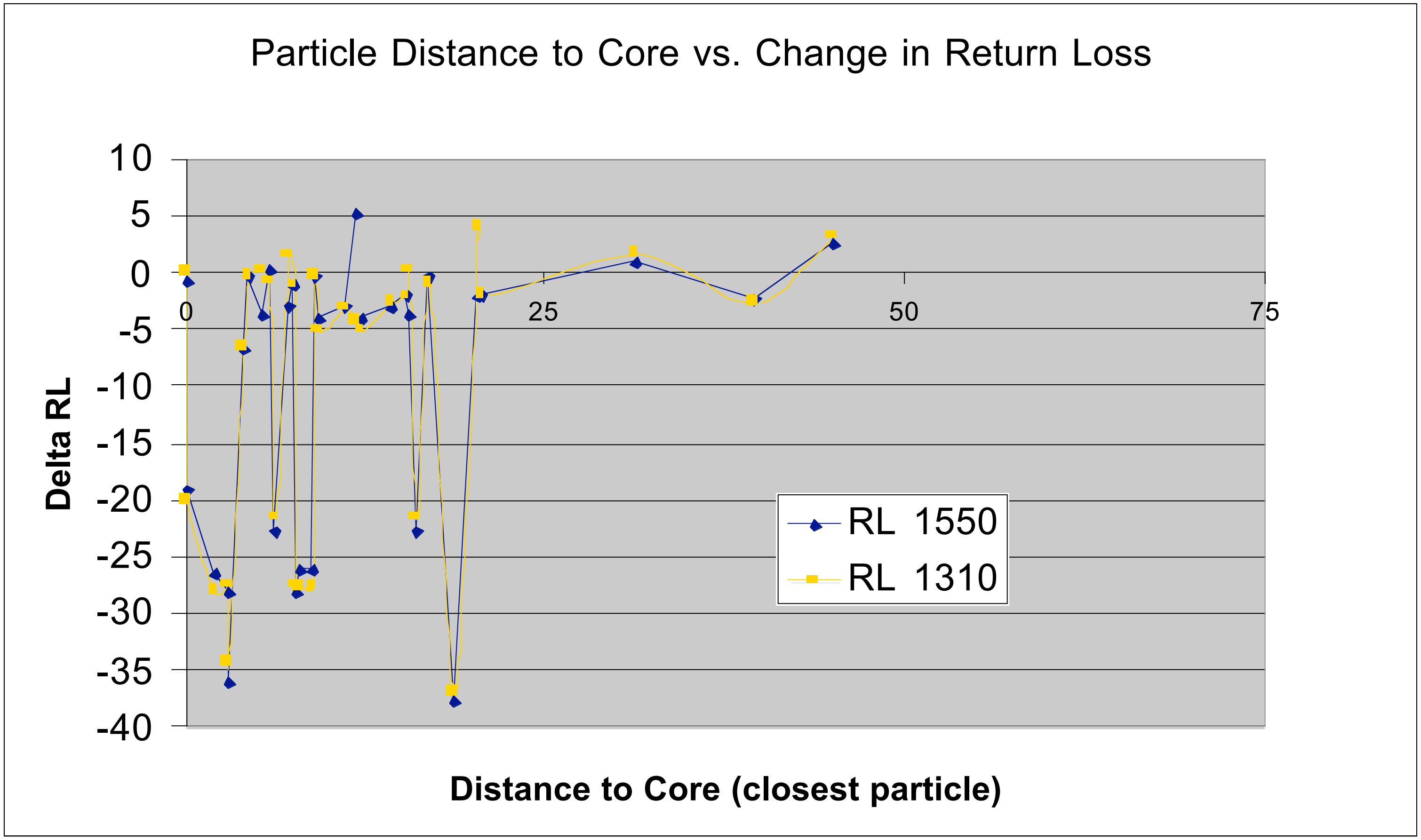
Fiber Optic Signal Performance Project



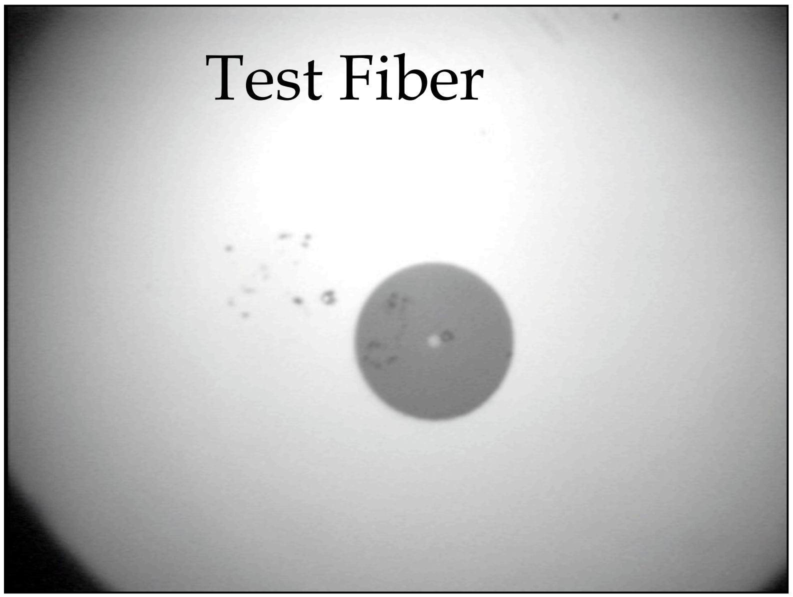

Particle Distance to Core vs. Change in Insertion Loss
3
2.5
2
1.5
1
0.5
0
-0.5
-1
Delta IL
0
25
50
75
Distance to Core (closest particle)
IL 1550
IL 1310

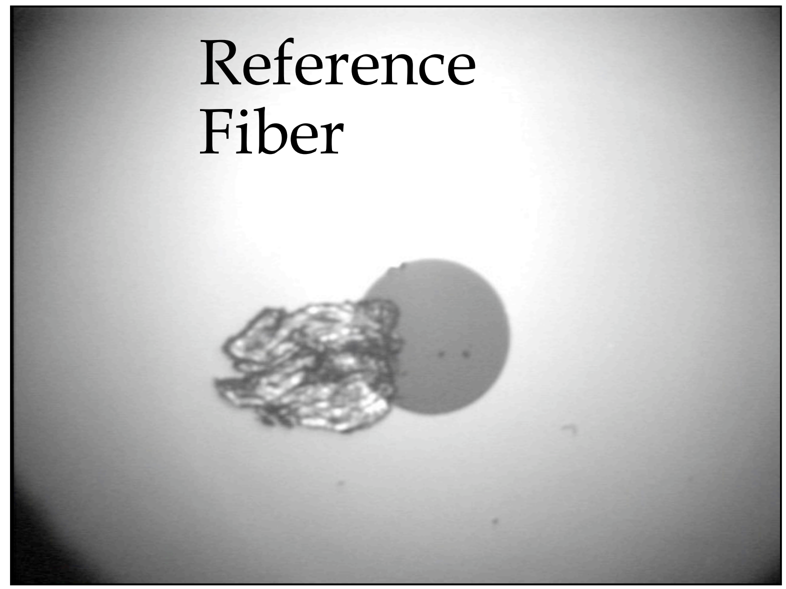
Fiber Optic Signal Performance Project




Particle Distance to Core vs. Change in Return Loss
Delta RL
10
5
0
-5
-10
-15
-20
-25
-30
-35
-40
0
25
50
75
RL 1550
RL 1310
Distance to Core (closest particle)

Fiber Optic Performance Project





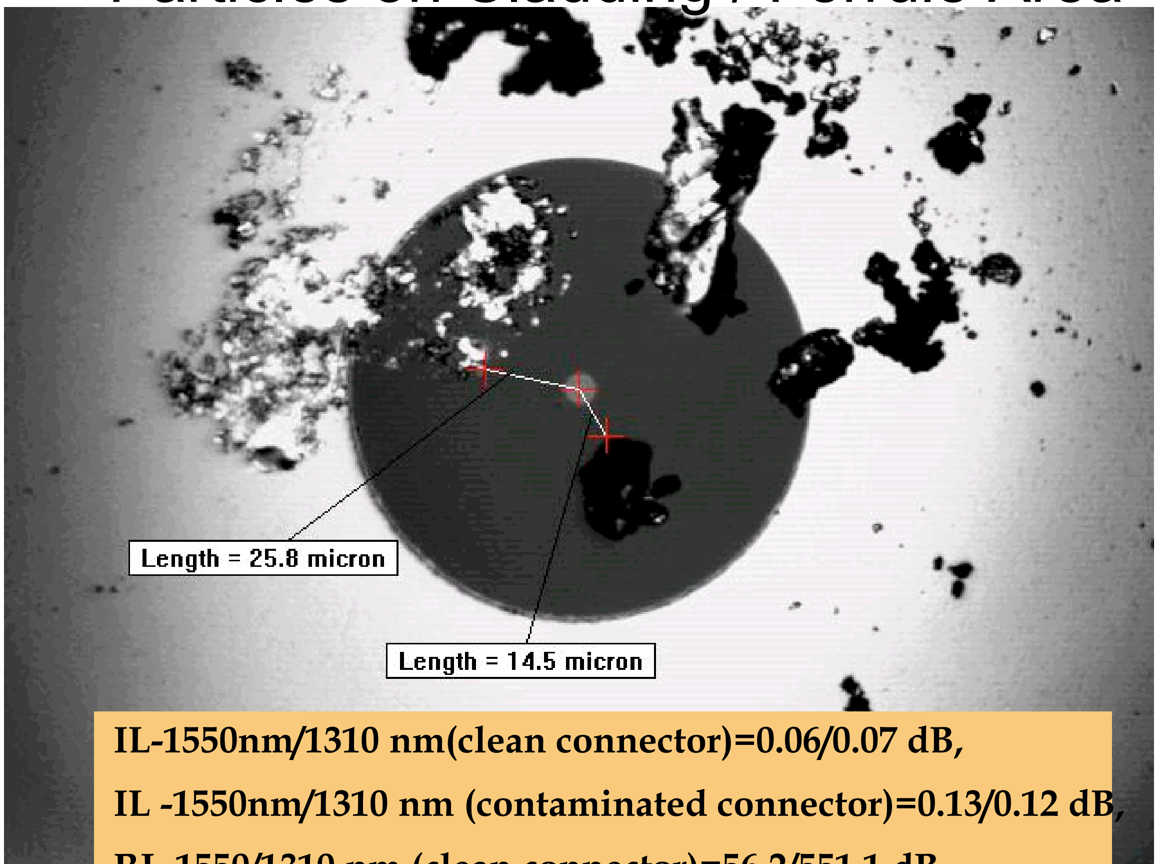
IL: 1550/1310 (contaminated)=0.25/0.20 dB

RL:1550/1310 (contaminated)=36.1/36.2 dB

IL: 1550/1310 (clean)=0.23/0.18 dB

RL=1550/1310 (clean)=58.7/57.8 dB

Particles on Cladding / Ferrule Area



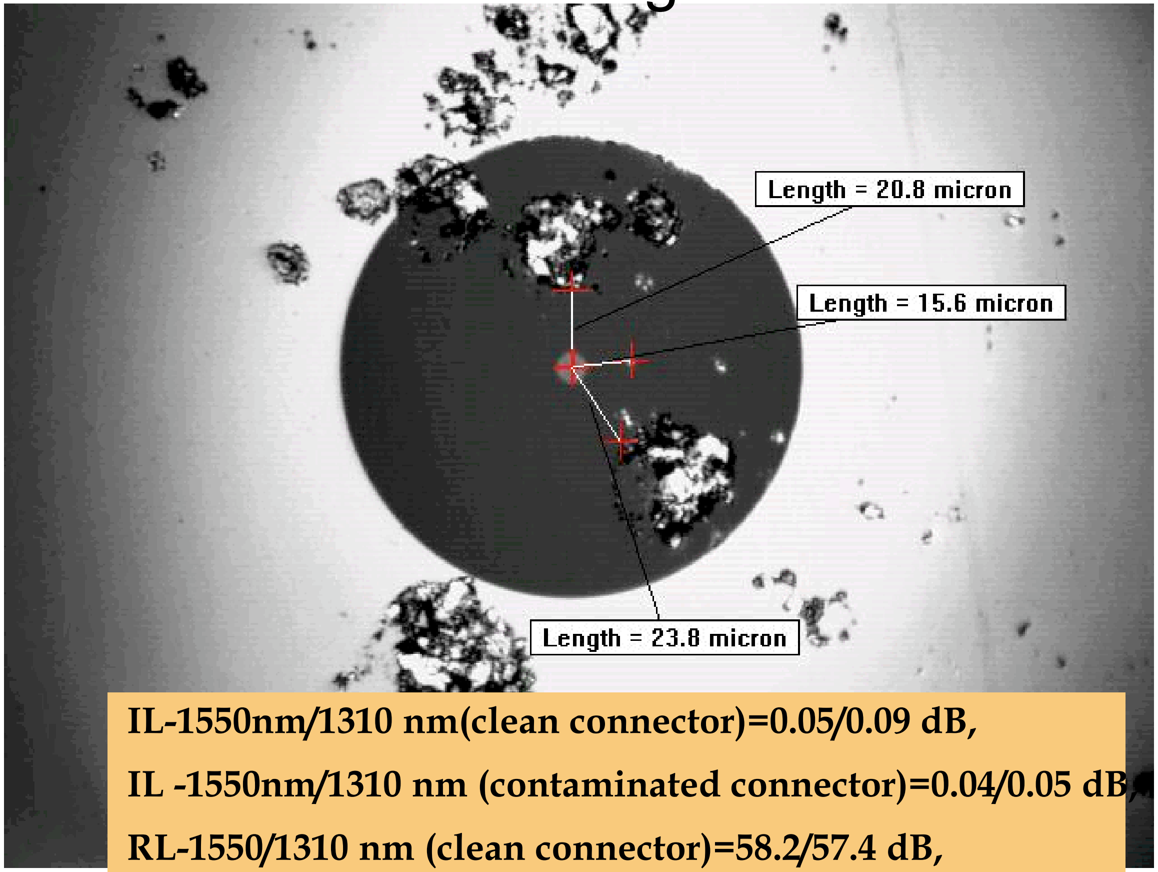
IL-1550nm/1310 nm(clean connector)=0.06/0.07 dB,

IL -1550nm/1310 nm (contaminated connector)=0.13/0.12 dB,

RL-1550/1310 nm (clean connector)=56.2/551.1 dB,

RL-1550nm/1310 nm (contaminated connector)=56.4/55.4 dB

Particles on Cladding / Ferrule Area



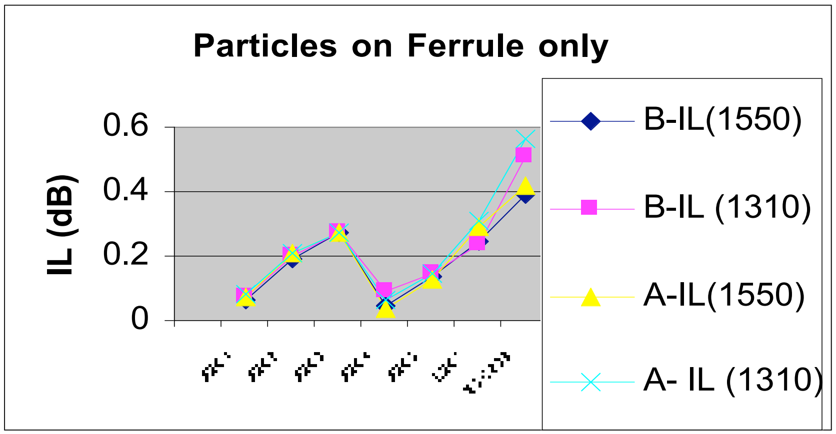
IL-1550nm/1310 nm(clean connector)=0.05/0.09 dB,

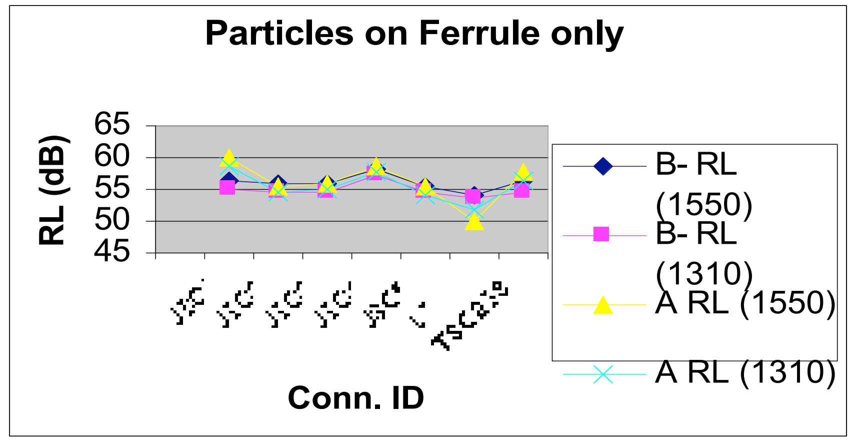
IL -1550nm/1310 nm (contaminated connector)=0.04/0.05 dB,

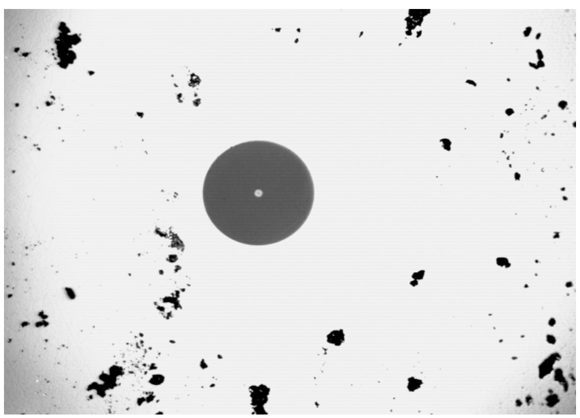
RL-1550/1310 nm (clean connector)=58.2/57.4 dB,

RL-1550nm/1310 nm (contaminated connector)=57.3/56.4 dB

Particles on Ferrule



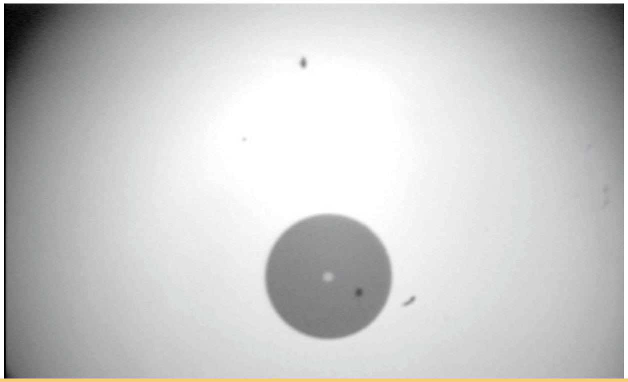




- **IL-1550nm/1310nm (clean)=0.19 dB/0.21 dB**
- **IL-1550 nm/1310nm (contaminated)=0.21dB/021 dB**
- **RL-1550nm/1310nm (clean)=55.7 dB/54.5 dB**
- **RL -1550nm/1310nm (contaminated)=55.6 dB/54.5 dB**

- The carbon/dust particles located at the distance less than 25 um from the core can resulted in the a significant increasing of the IL (0.5-1 dB) and decreasing a RL (20-30 dB change)
- Based on our results the carbon/dust particles located at the closest distance greater than 25 um from the core have minimum impact at IL (< 0.02 dB) and RL (< 4dB)
- Removable contamination can be redistributed during the mating/demating operation.The removable contamination is not acceptable.
- The thickness/hardness of the particle can be critical. The further investigation is required

- **Critical parameters:**
  - Distance the particle from the core
  - Size of the particle
  - Type of the particle
  - Hardness of the particle/thickness

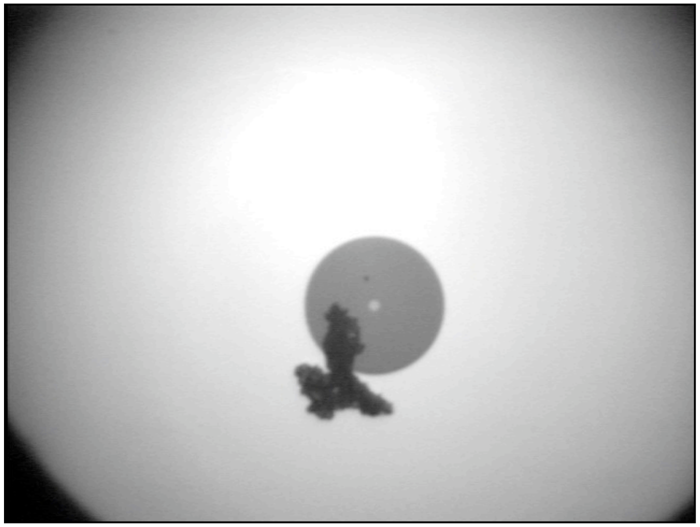


IL=1550/1310 (contaminated)=0.14/0.15 dB

RL=1550/1310 (contam.)=59.6/58.5 dB

IL=1550/1310 (clean)=0.11/0.11 dB

RL=1550/1310 (clean)=58.6/56.8 dB



IL=1550/1310 (contaminated)=16.77/18.08 dB

RL=1550/1310 (contaminated)=20.3/20.8 dB

IL=1550/1310 (clean)=0.11/0.05 dB

RL=1550/1310(clean)=58.0/57.6dB

Potential effects of the particles on the Optical Loss:

- Particles at the core:
  - Light scattering;
  - Damaging of the core (pits/scratches)
  - Reflections (Index reflection change)
  - Core misalighnment, Angle misalighnment;
  - Endface separation (air gap)

- Particles at the cladding layer and the ferrule
  - Damaging of the cladding area and the ferrule (pits/scratches)
  - Core misalighnment, Angle misalighnment;
  - Endface separation (air gap)

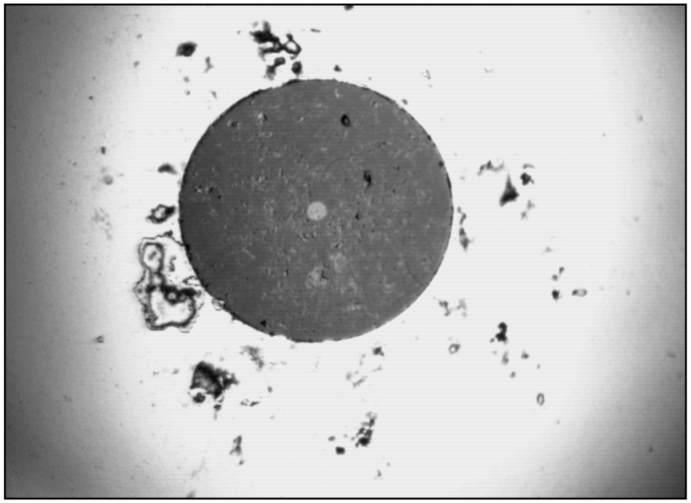


Fig. 1

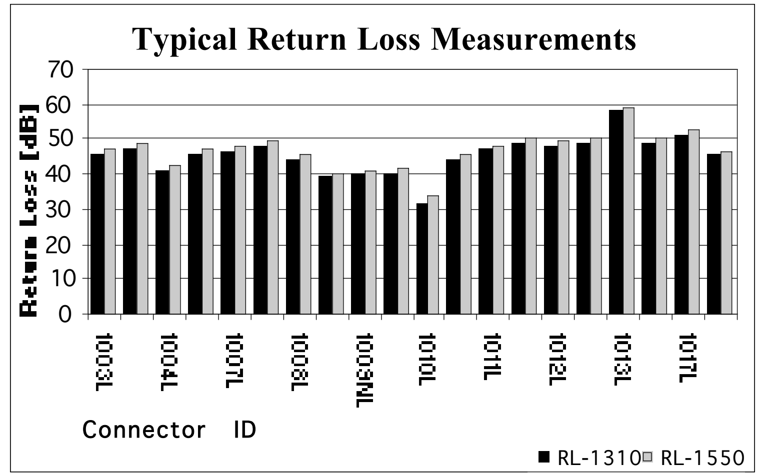
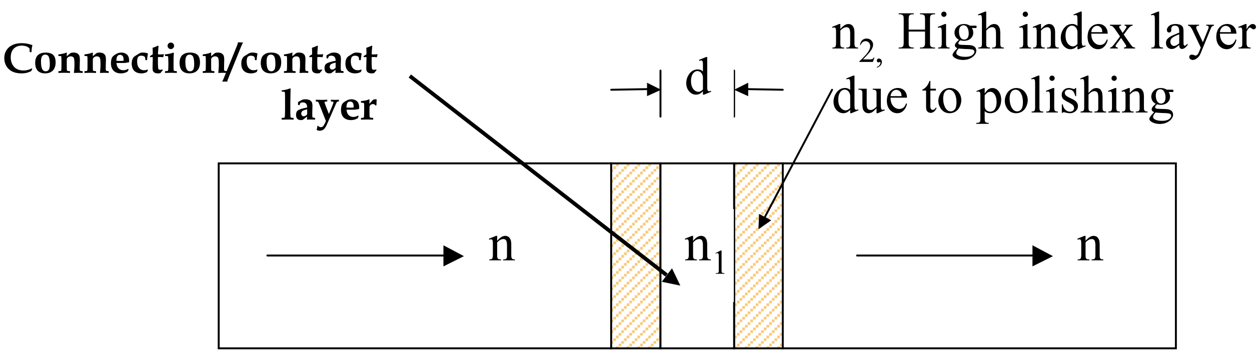


Fig. 2

Fig. 1 Typical endface of the connector used in standard manufacturing environment

Fig. 2 RL measurements show performance degradation of aged connectors in a manufacturing environment.



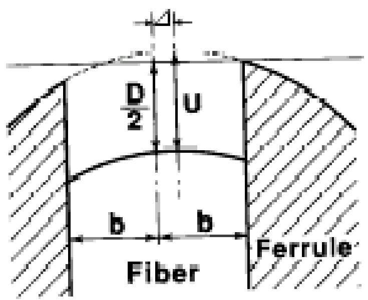
$$RL(dB) = 10\log\left[2R\left(1 - \cos\left(\frac{4\pi n_1}{\lambda} d\right)\right)\right]$$

$$R = (r_1^2 + r_2^2 + 2r_1 r_2 \cos\delta)\big/(1 + r_1^2 r_2^2 + 2r_1 r_2 \cos\delta)$$

$$r_1 = \frac{n_0 - n_2}{n_0 + n_2}, \quad r_2 = \frac{n_2 - n_1}{n_2 + n_1}, \quad \delta = \frac{4\pi}{\lambda} n_2 h$$

The following factors have been taken into the consideration:

- **Higher index layer due to the polishing**
  - **n=1.4677 (SMF-28), n_1=1, n_2 = 1.476, h=20nm**
- **Geometric parameters (radius of the curvature, apex offset and fiber undercut)**
- **Increasing of the roughness of the surface (‘ contamination” layer with thickness 1-4 nm)**



$$D/2 = (U + b\square/R)\square b^2/(2R)$$

T. Shintaku, “ J. Lightwave Technology, V.11, #2, 241-247

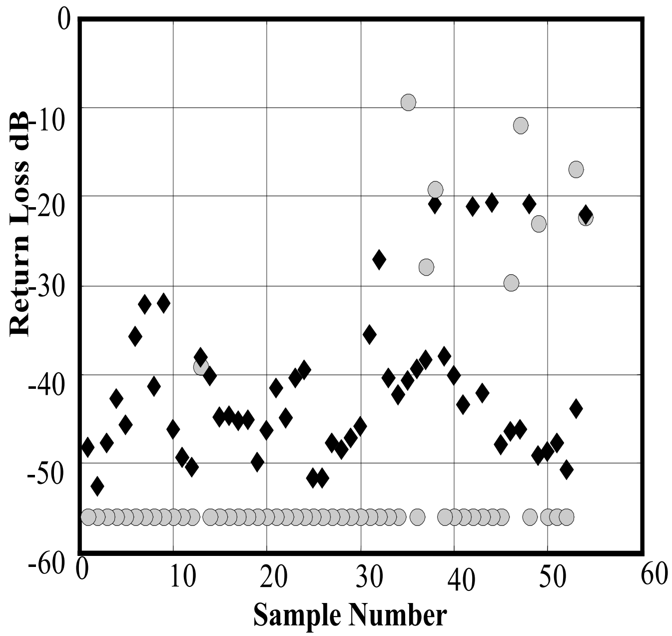


Fig.1

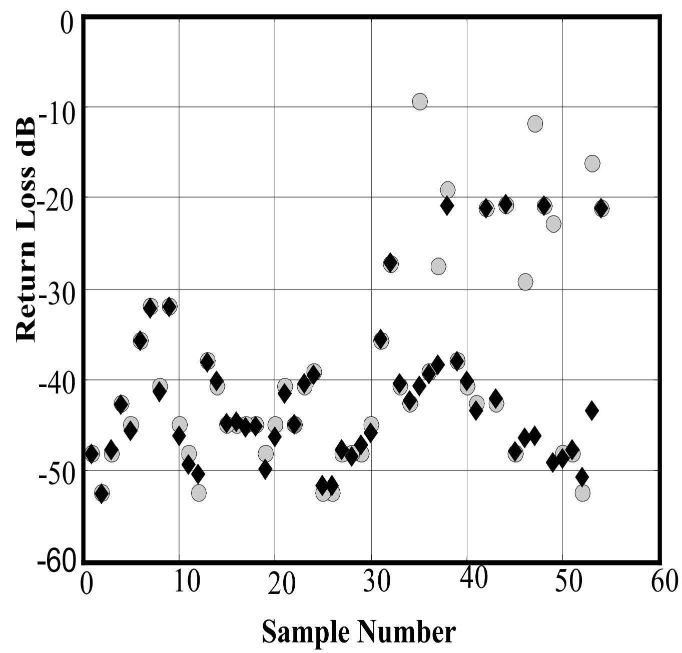
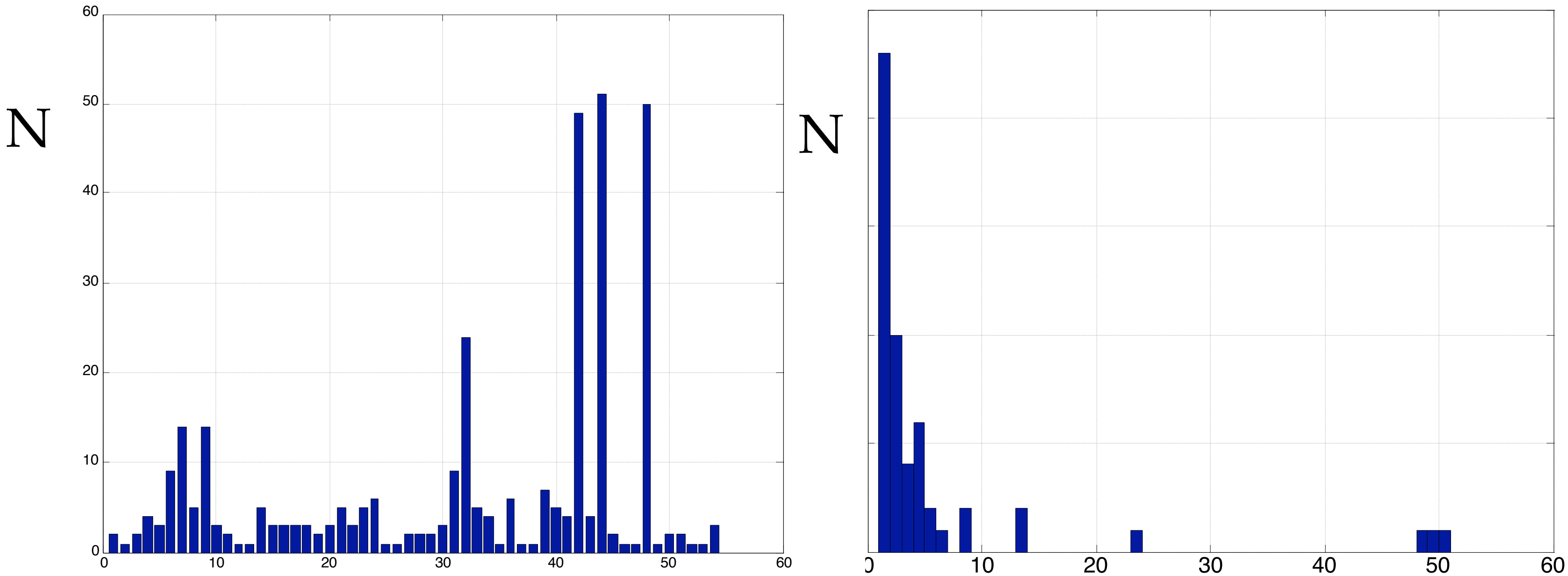


Fig.2

Fig. 1 Measured (diamonds) and calculated (circles) RL at 1310 nm with air gap caused by geometric factors

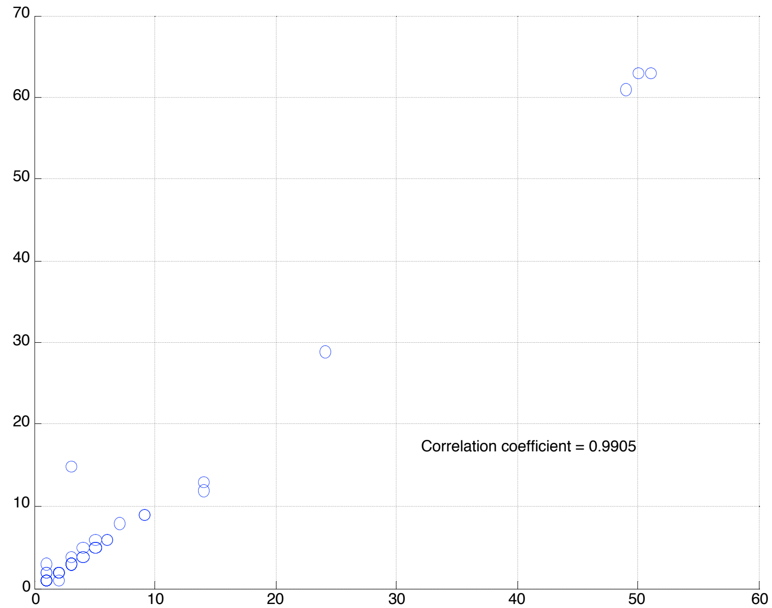
Fig.2 Measured (diamonds) and calculated (circles) RL at 1310 nm with an air gap caused by geometric factors and connection (contact) layer



Thickness of connection/contact layer, nm

The histogram of the gap size due to the connection/contact layer. The wavelength is 1310 nm

Estimated thickness of contact layer, 1550 nm



Estimated thickness of contaminated layer, 1310 nm

The correlation between estimated thickness of contact/connection layer for 1310 nm and 1550 nm. The high correlation value 0.99 supports the model that uses a contact/connection layer to explain RL

Oil Contamination

Cleaning&

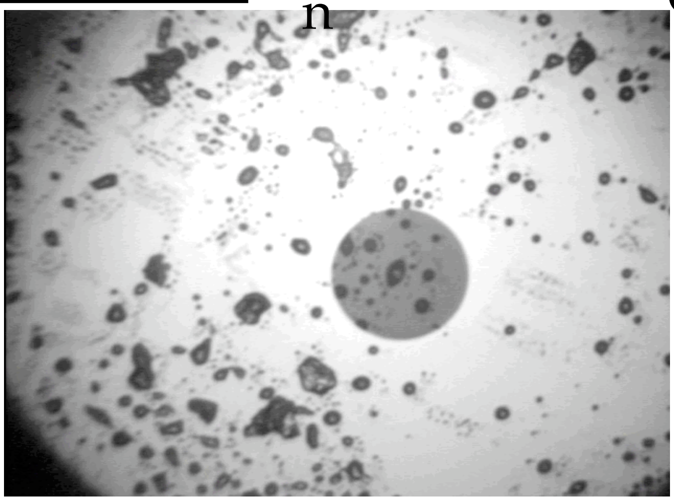
Visual Inspection

→ Test Connector, Oil Application

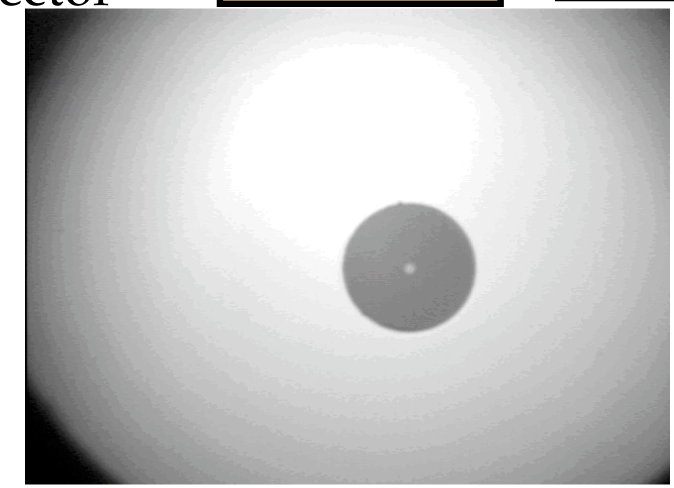
→ Mating with clean Ref. Connector

→ IL& RL, 2 matings

→ Visual Inspection

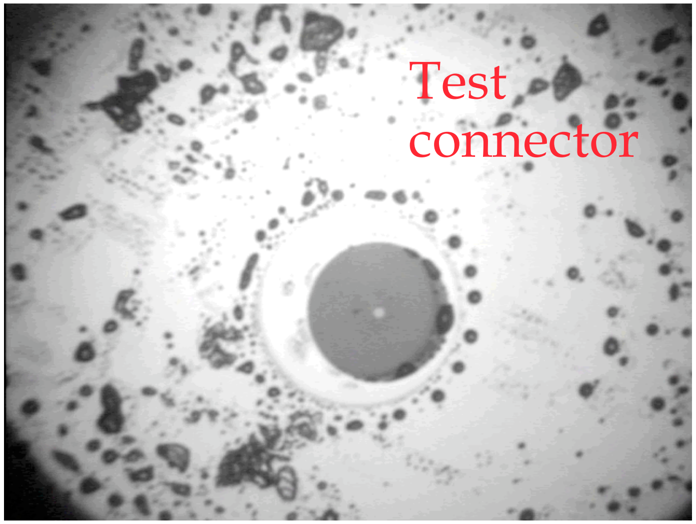


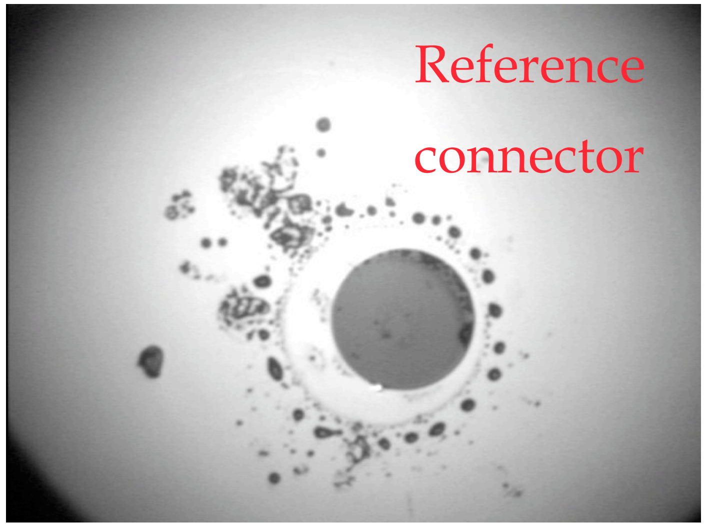
Oil contaminated connector before the mating



Clean Reference connector before the mating

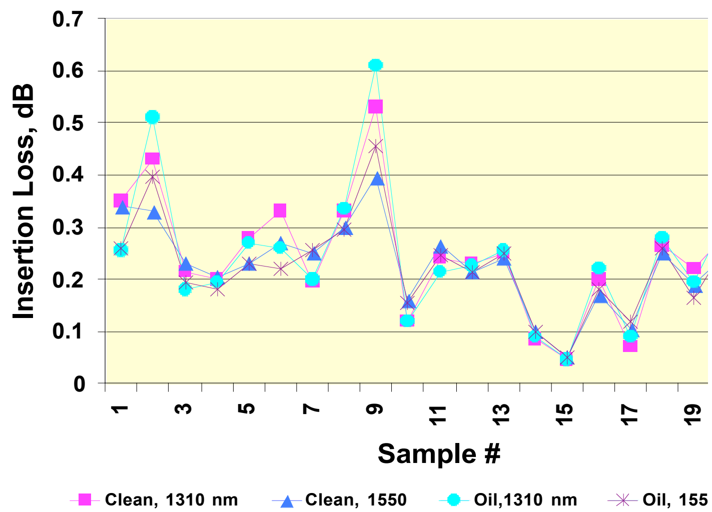
Oil Contamination

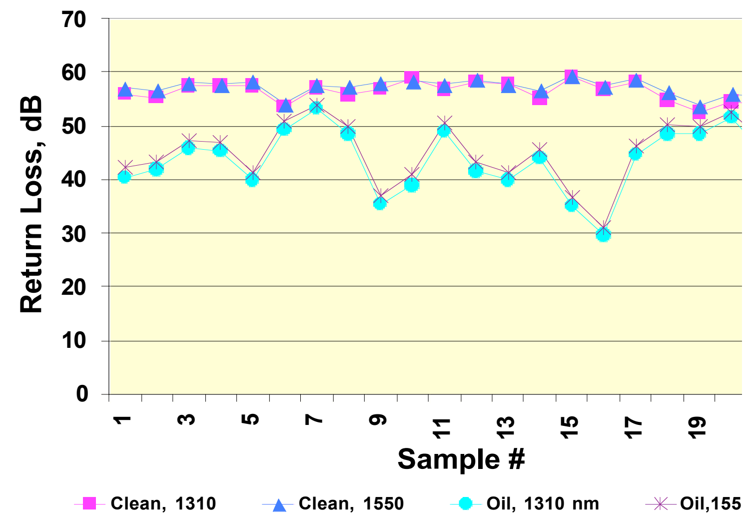


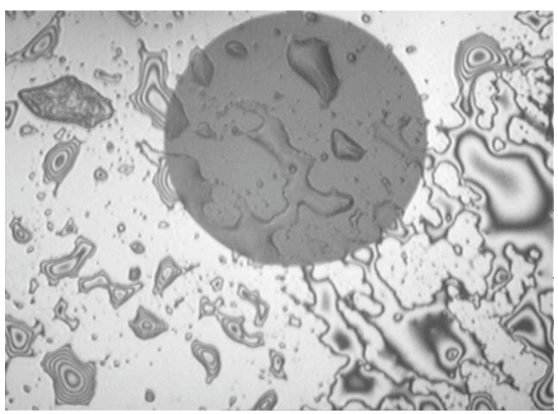


- IL (clean connector)-1550nm/1310nm=0.22dB/0.27dB;
- IL (contaminated connector)-1550nm/1310 nm/=0.23dB/0.27dB;
- RL (clean connector)-1550nm/1310nm=58.1dB/56.9dB;
- RL (contaminated connector)-1550nm/1310nm=41.2/39.8dB

Oil Contamination

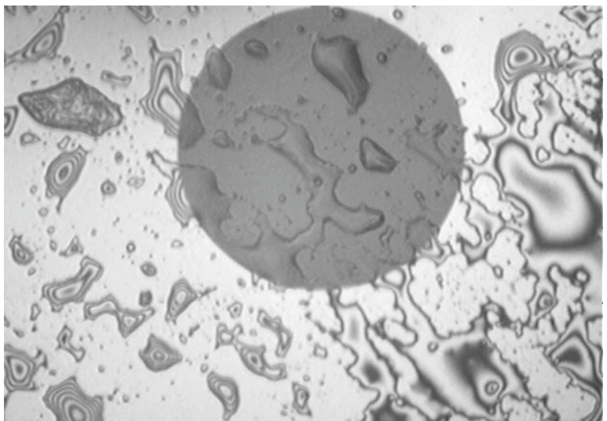


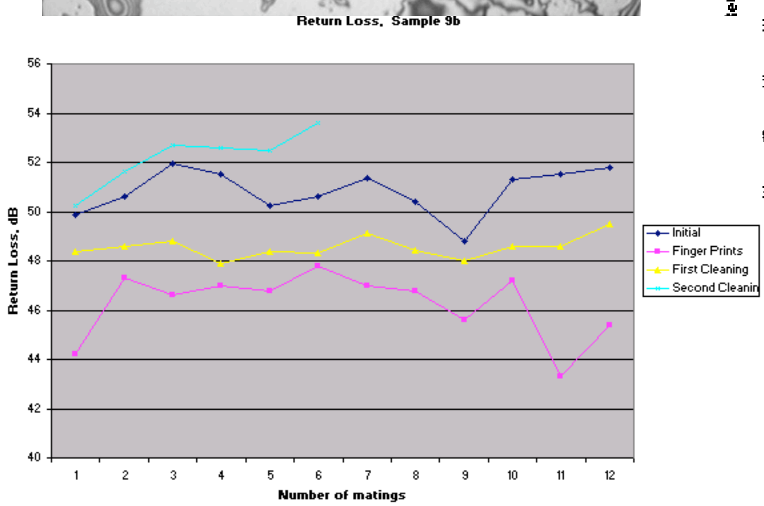


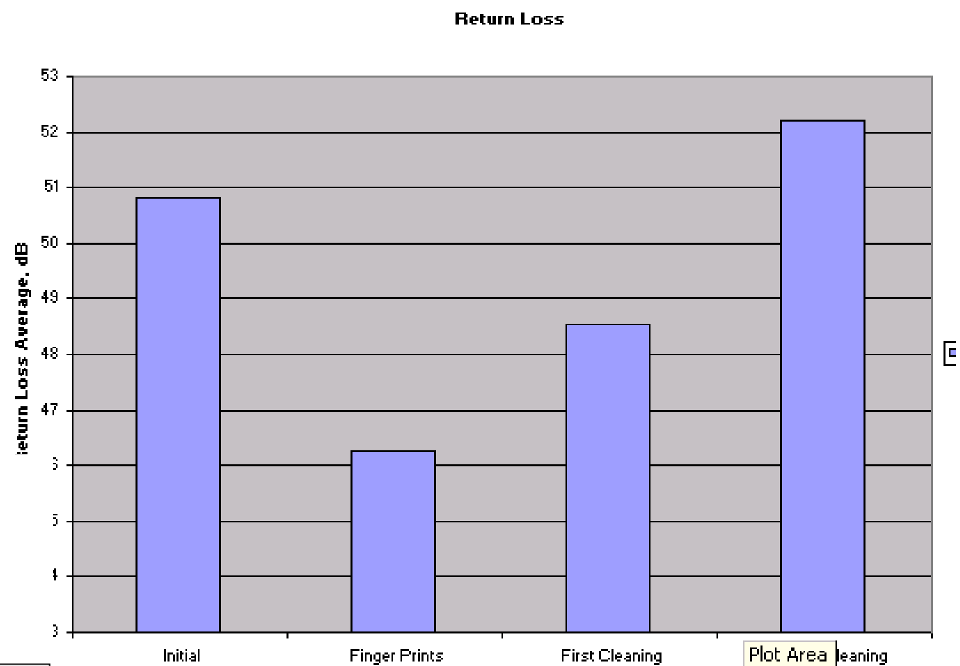


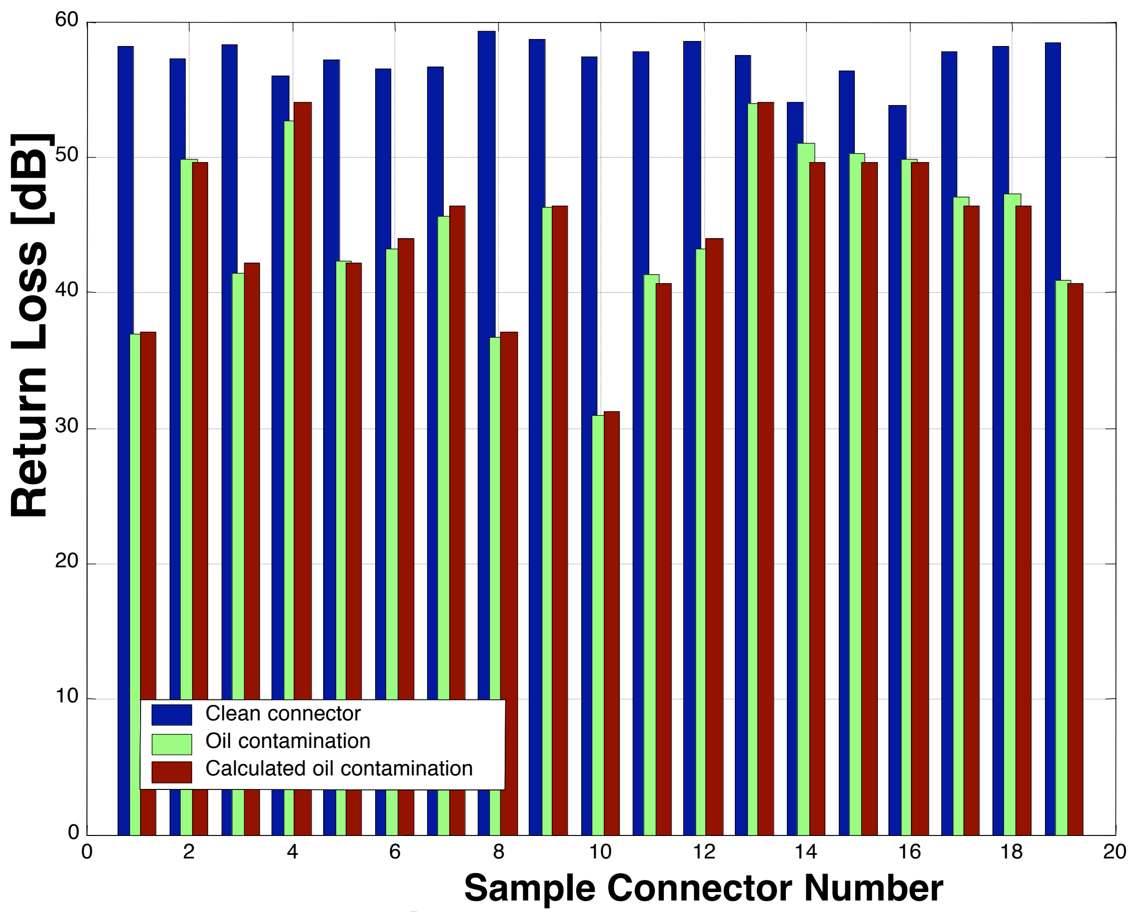
- Oil application resulted in significant changes of RL:
- RL decreased from 56.3 dB to 43.6 dB (wavelength-1310nm) and from 57.2dB to 45.1 dB (wavelength 1550 nm)

Oil Contamination



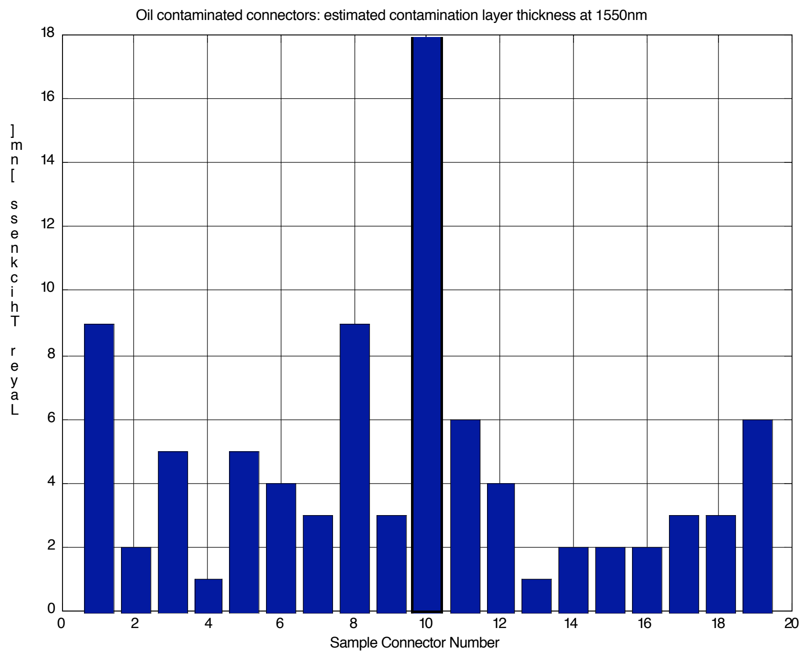


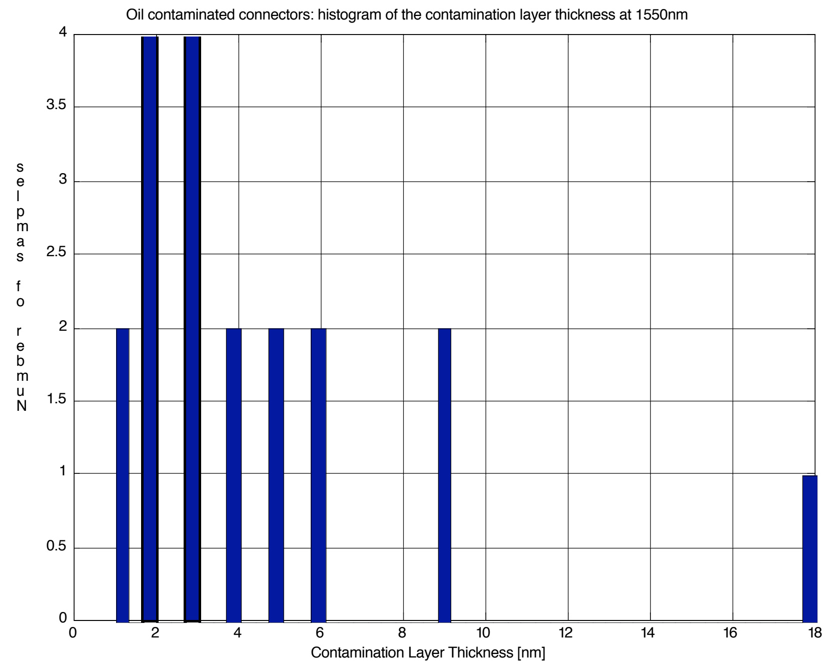




Oil contaminated connectors: RL at 1550nm

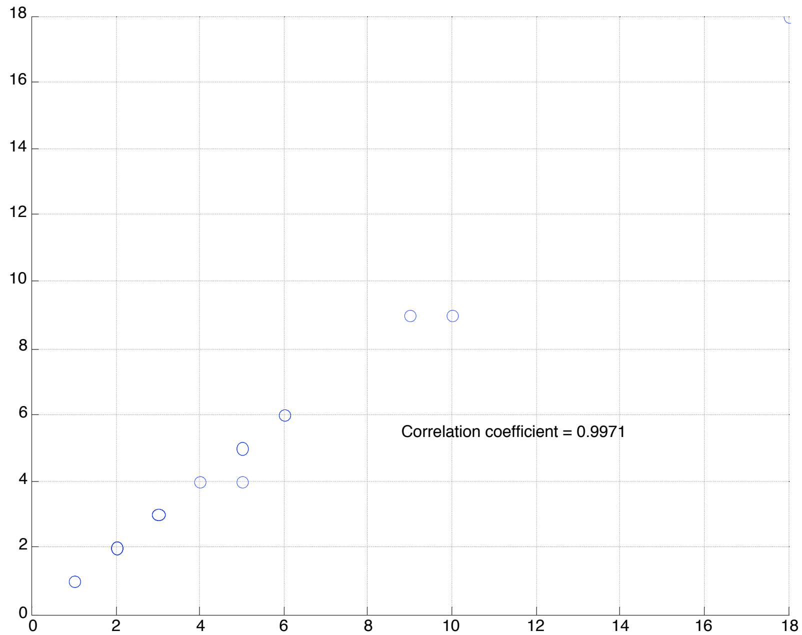
Fiber Optic Signal Performance Project





- Oil contamination. More than 90 % of the results can be explained by the thickness of oil layer 1-8 nm

Fiber Optic Signal Performance Project

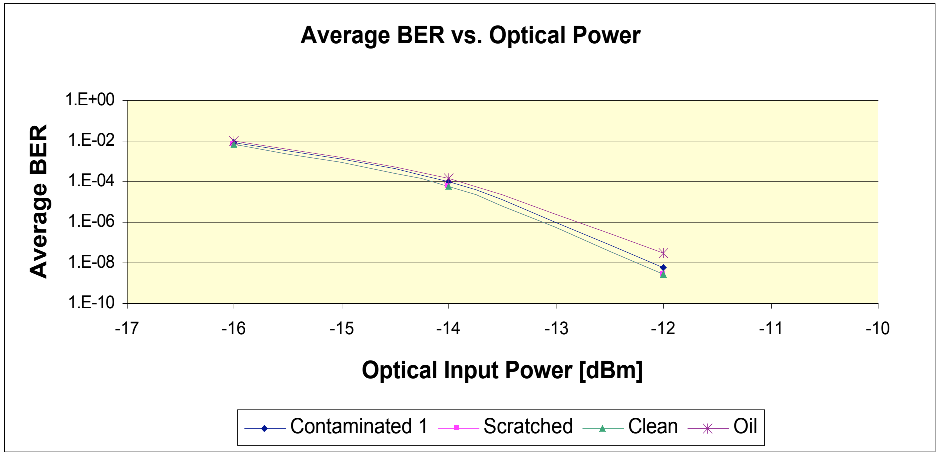


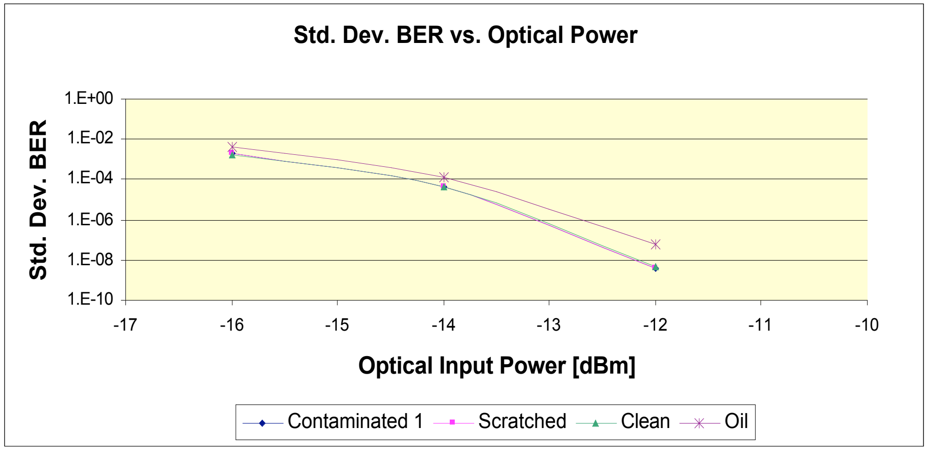
Estimated thickness of oil layer, 1550 nm

Estimated thickness of oil layer, 1310 nm

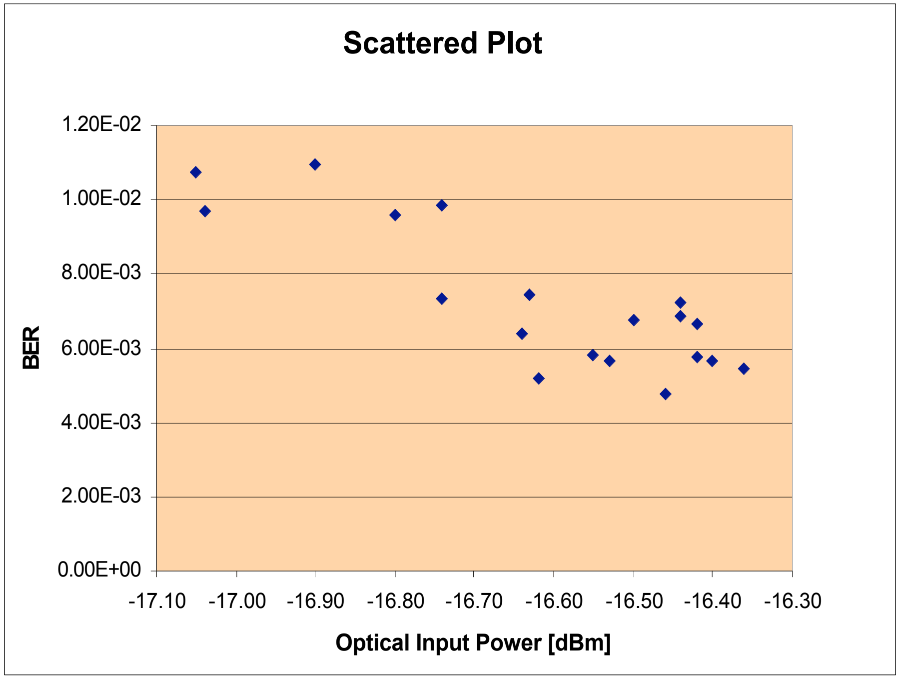
The correlation between estimated thickness of contact/connection layer for 1310 nm and 1550 nı
The high correlation value 0.99 supports the model that uses an oil layer to explain RL

BERT Results

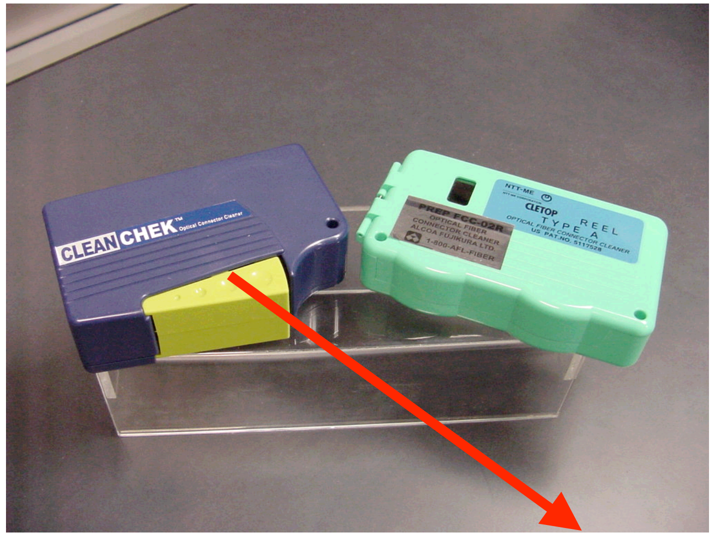




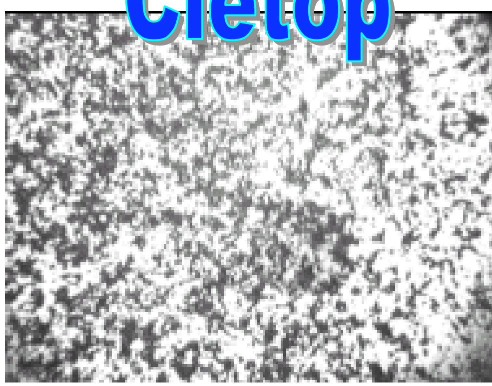
BERT Results Analysis



- The BERT results and Optical Input Power are anti-correlated (correlation coefficient is -0.84)

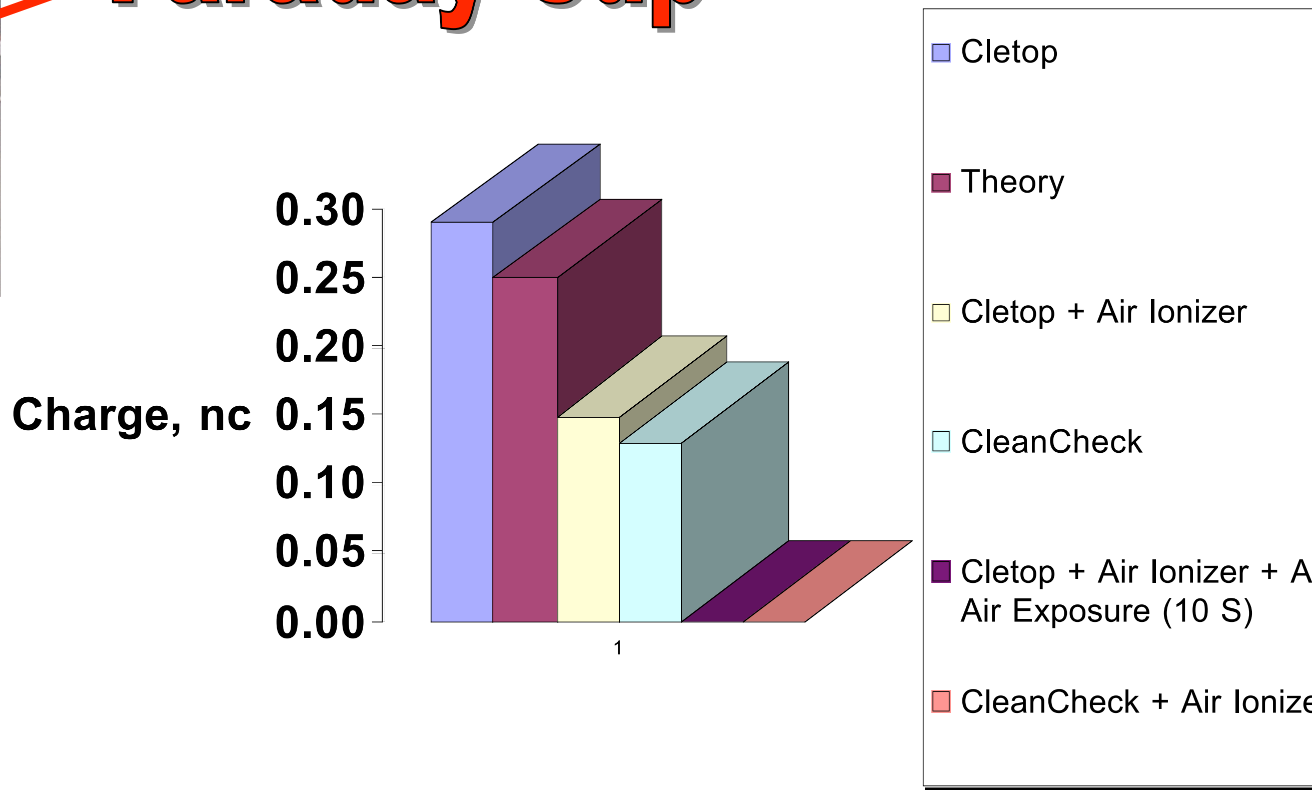


Cletop



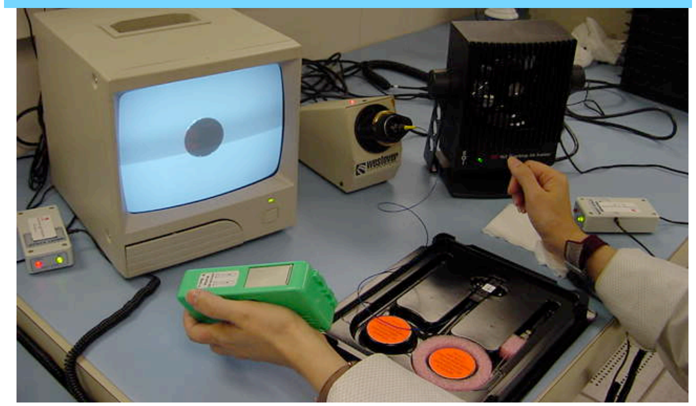
Optipop

Faraday Cup

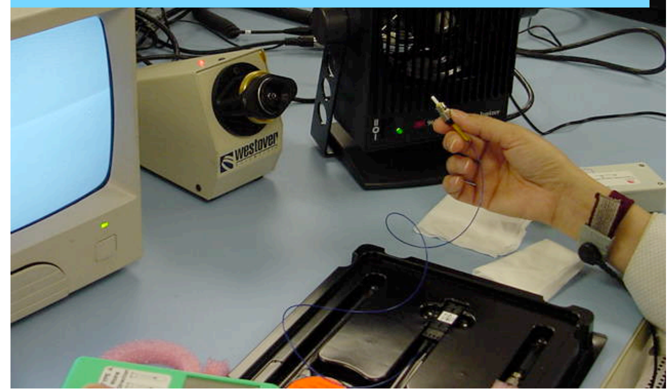

Charge, nc
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1
Cletop
Theory
Cletop + Air Ionizer
CleanCheck
Cletop + Air Ionizer + A
Air Exposure (10 S)
CleanCheck + Air Ionize

Fiber Optic Signal Performance Project

Application of air ionizers

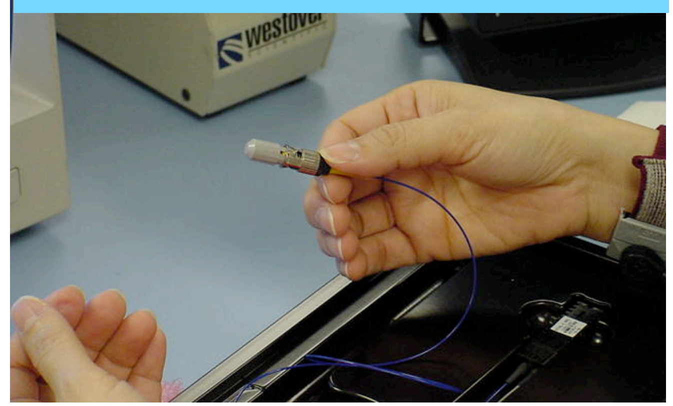


Exposure to ionized air



Optipop cassette

Protection with a dust cap



Conclusions

- The influence of scratches, particles and oil contamination on optical performance of SC connectors was investigated
- Scratches (2um or less), within MFD, has no impact on IL can degrade the RL
- The level of RL degradation depends on the size (width and depth) and, the number of the scratches crossing MFD
- Polishing scratches outside the fiber MFD have no impact on IL & RL

Conclusions

- Particles on the core were resulted in catastrophic failures while the presence the particles on the ferrule does not show any degradation on its performance;
- Future studies will investigate more on the effects of the particles when they are locate at the cladding layer as well as focusing on particle size, quantity and different particle types
- Application of oil contamination resulted in significant degradation of RL (10-12 dB) and didn’t result in any significant changes of IL;

Conclusions

- The BERT results were anti-correlated with Optical Input Power (coefficient of the correlation is -0.84)
- The development of the mathematical modeling for scratches/particles/oil contamination is the subject of the further research